High-Speed Networking Technologies

Switching

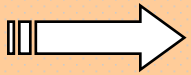
ATM and MPLS

Assoc.Prof.Dr.Chawalit Benjangkaprasert

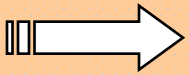
ATM (Asynchronous Transfer Mode)

ภาวะถ่ายโอนแบบไม่ประสานเวลา

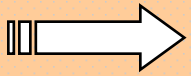
- ภาวะถ่ายโอนแบบไม่ประสานเวลา หรือ เอทีเอ็ม เป็นเทคโนโลยีการสื่อสารข้อมูลความเร็วสูงที่ใช้ในเครือข่ายโทรคมนาคม โดยข้อมูลจะถูกแบ่งเป็นเซลล์ขนาดเล็ก และส่งผ่านเครือข่ายแบบไม่ประสานเวลา



VOICE



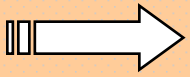
DATA



VIDEO

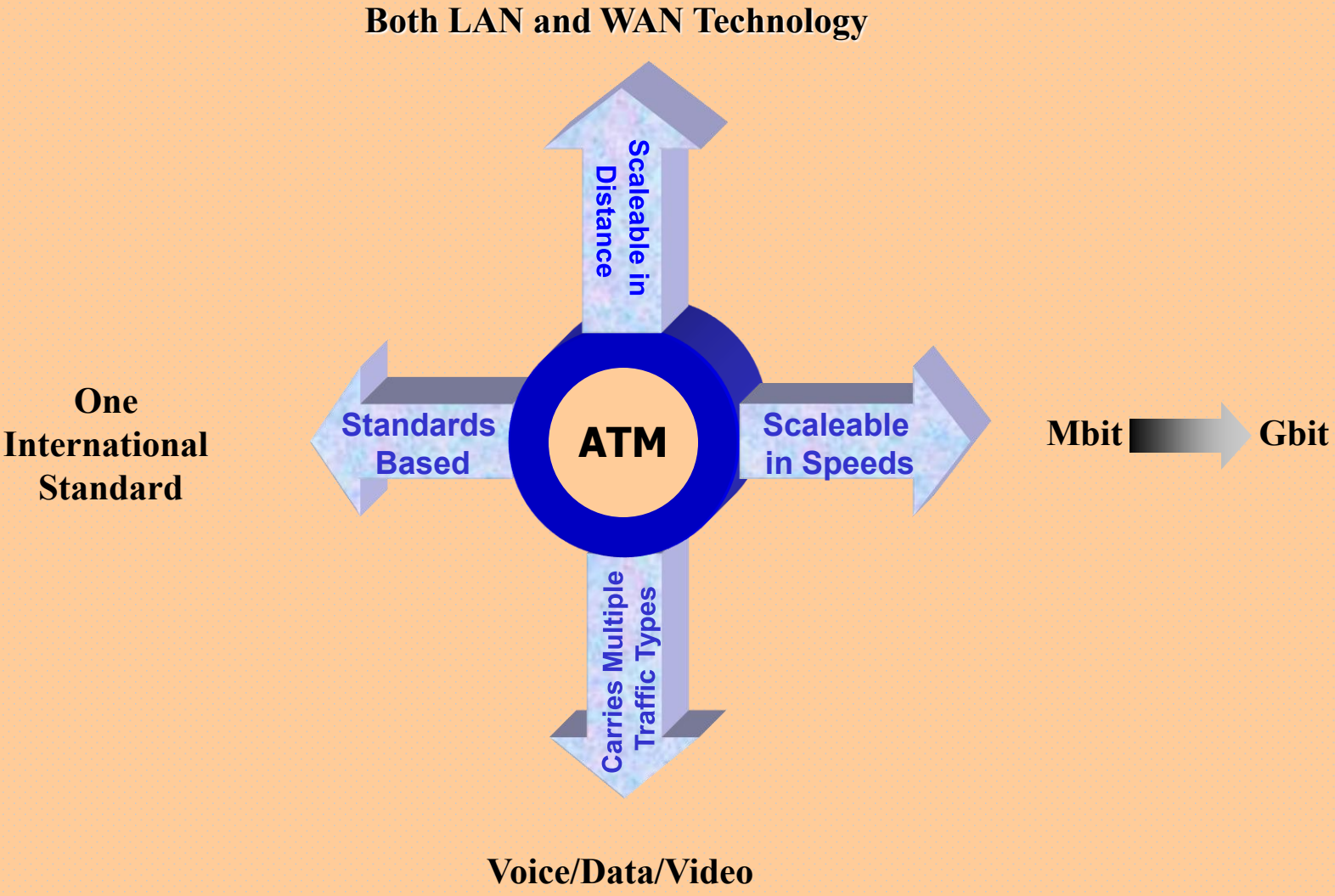
ATM (Asynchronous Transfer Mode)

- **ATM is the cell relay protocol designed by the ATM Forum and adopted by the ITU-T.**
- **ATM has been accepted universally as the transfer mode of choice for Broadband Integrated Services Digital Networks (BISDN) and others.**
- **ATM can handle any kind of information i.e. voice, data, image, text and video in an integrated manner.**



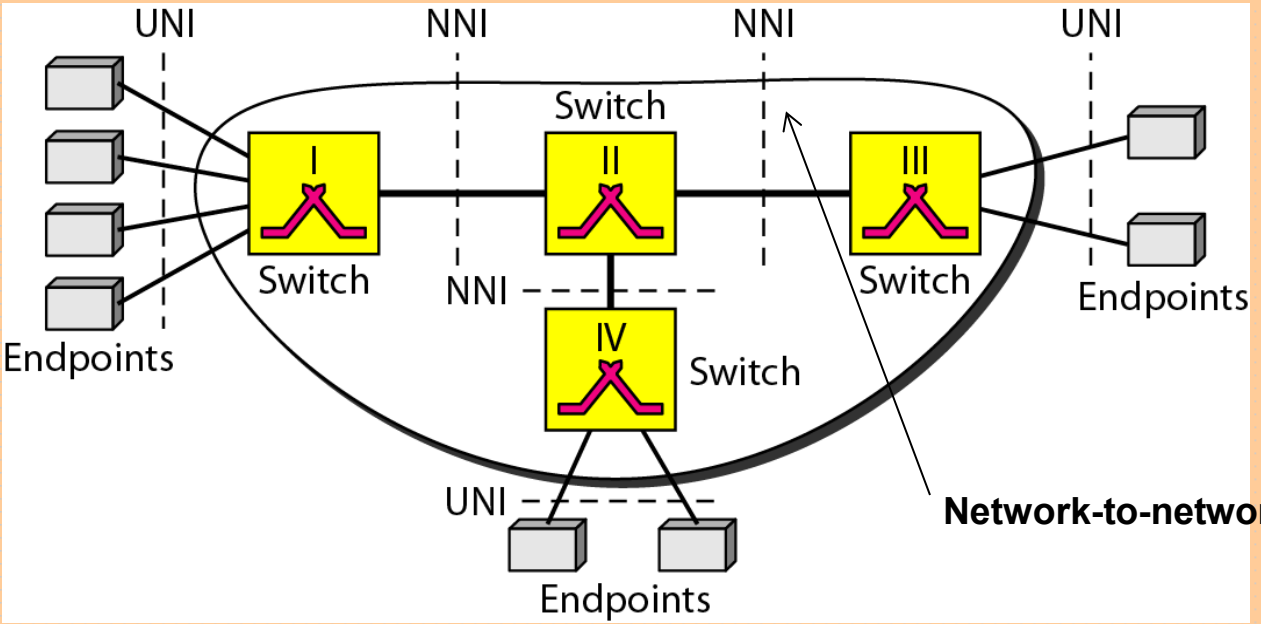
VOICE DATA VIDEO

Why Interest in ATM?



Architecture of an ATM network

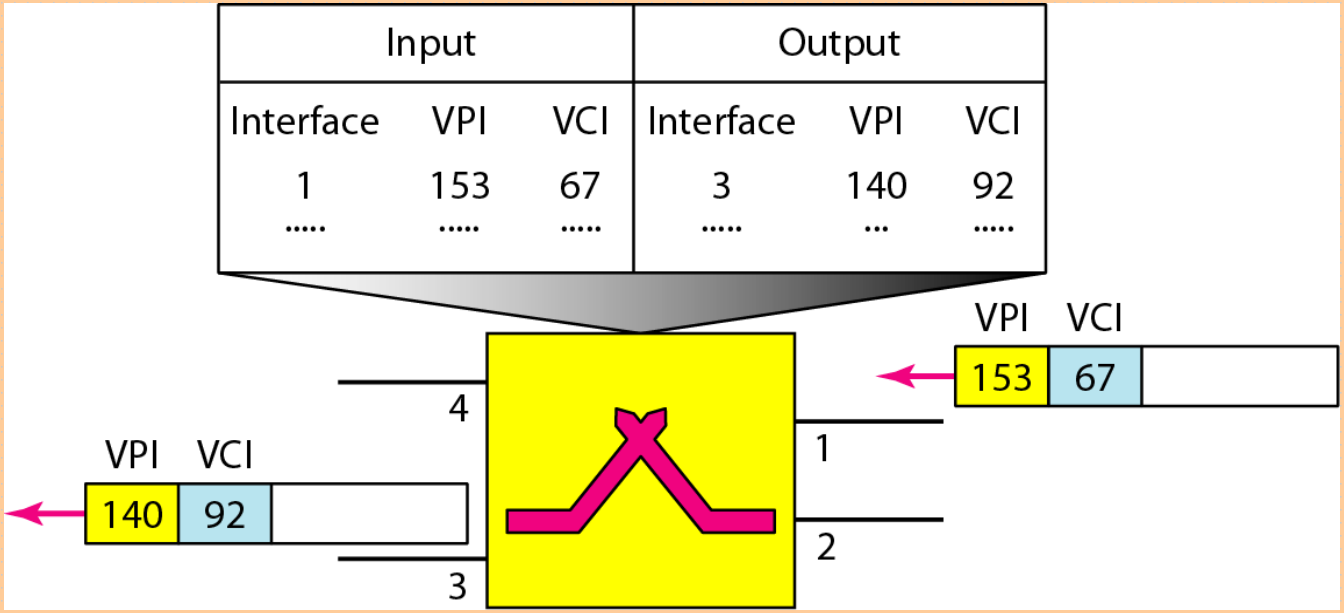
ATM is a cell-switched network. The user access devices, called endpoints, are connected through a user-to-network interface (UNI) to the switches inside the network.



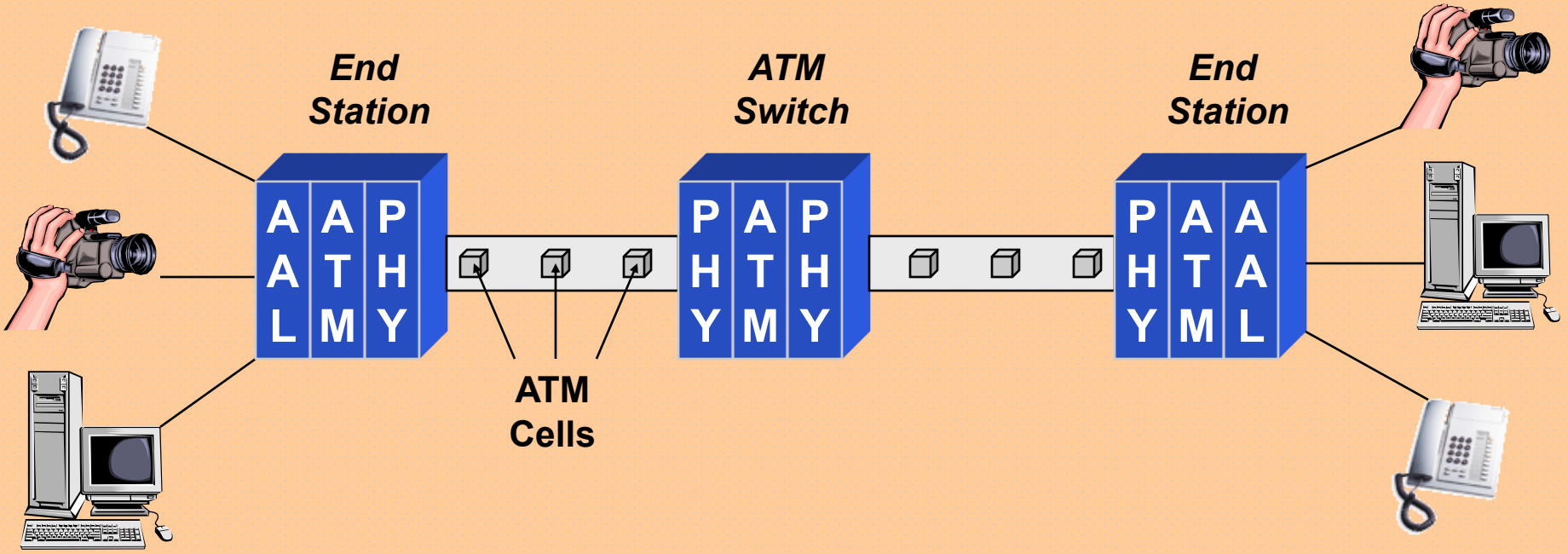
The switches are connected through network-to-network interfaces (NNIs).

Switching

Routing with a switch

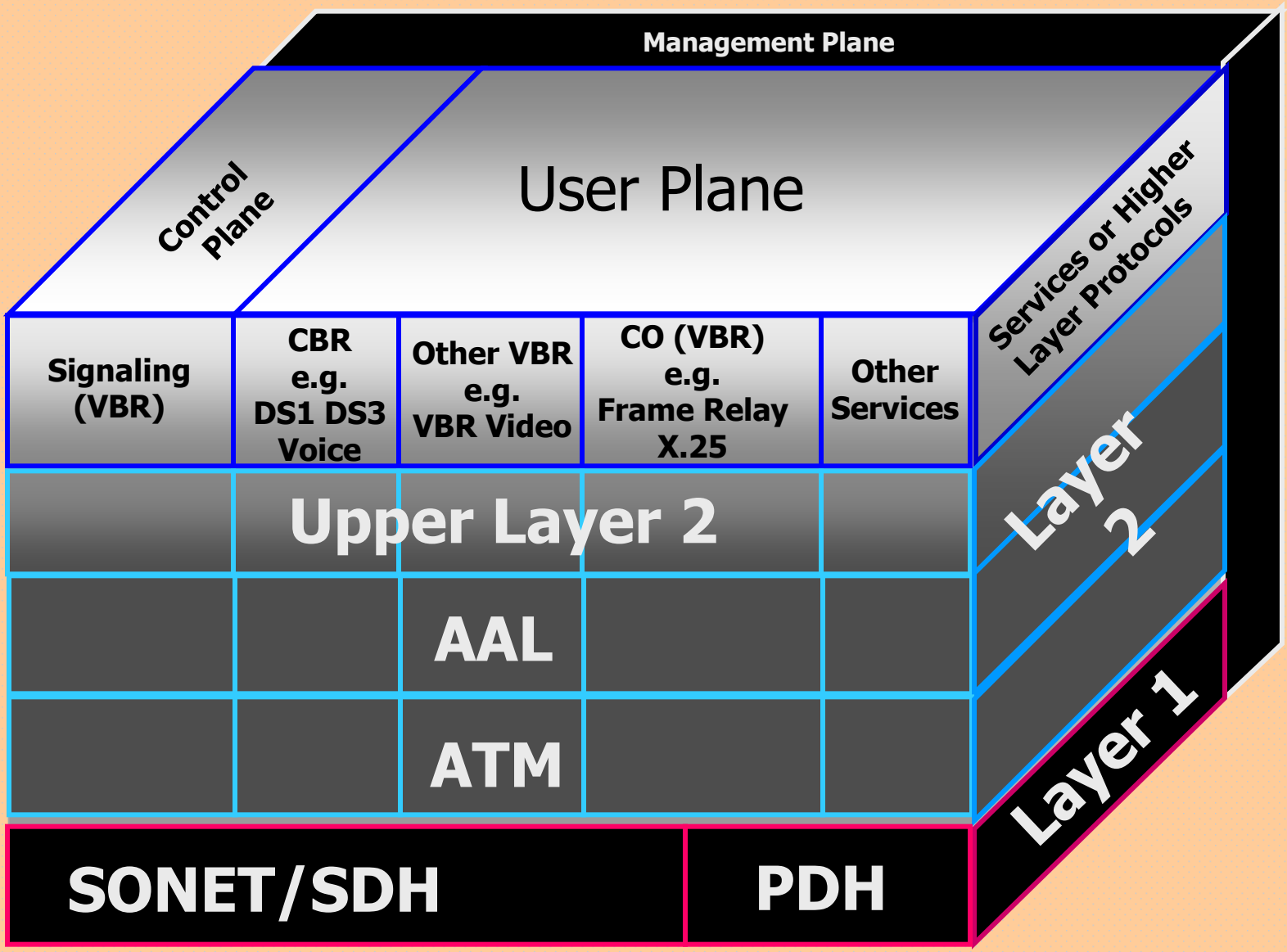


Functions of Layers



- **Application Adaptation Layer (AAL): Inserts/extracts information into 48 bytes payload**
- **ATM Layer: Adds/removes 5 bytes header to payload**
- **Physical Layer: Converts to appropriate electrical or optical format**

Broadband Protocol Model

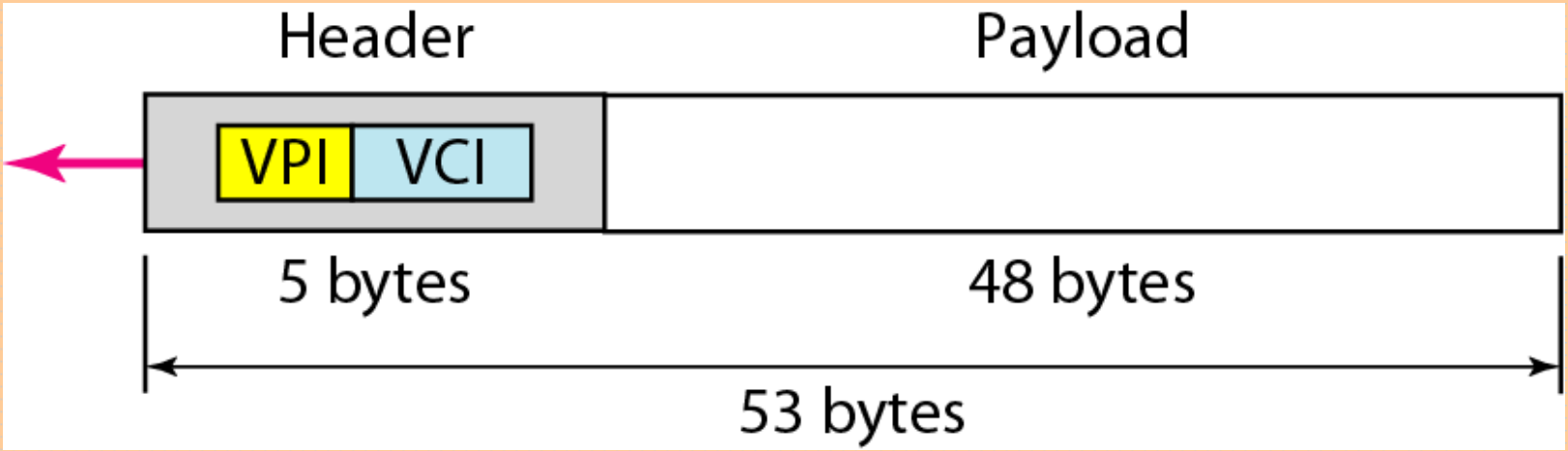


Synchronous Optical Network/Synchronous Digital Hierarchy, Plesiochronous Digital Hierarchy

Structure of an ATM Cell

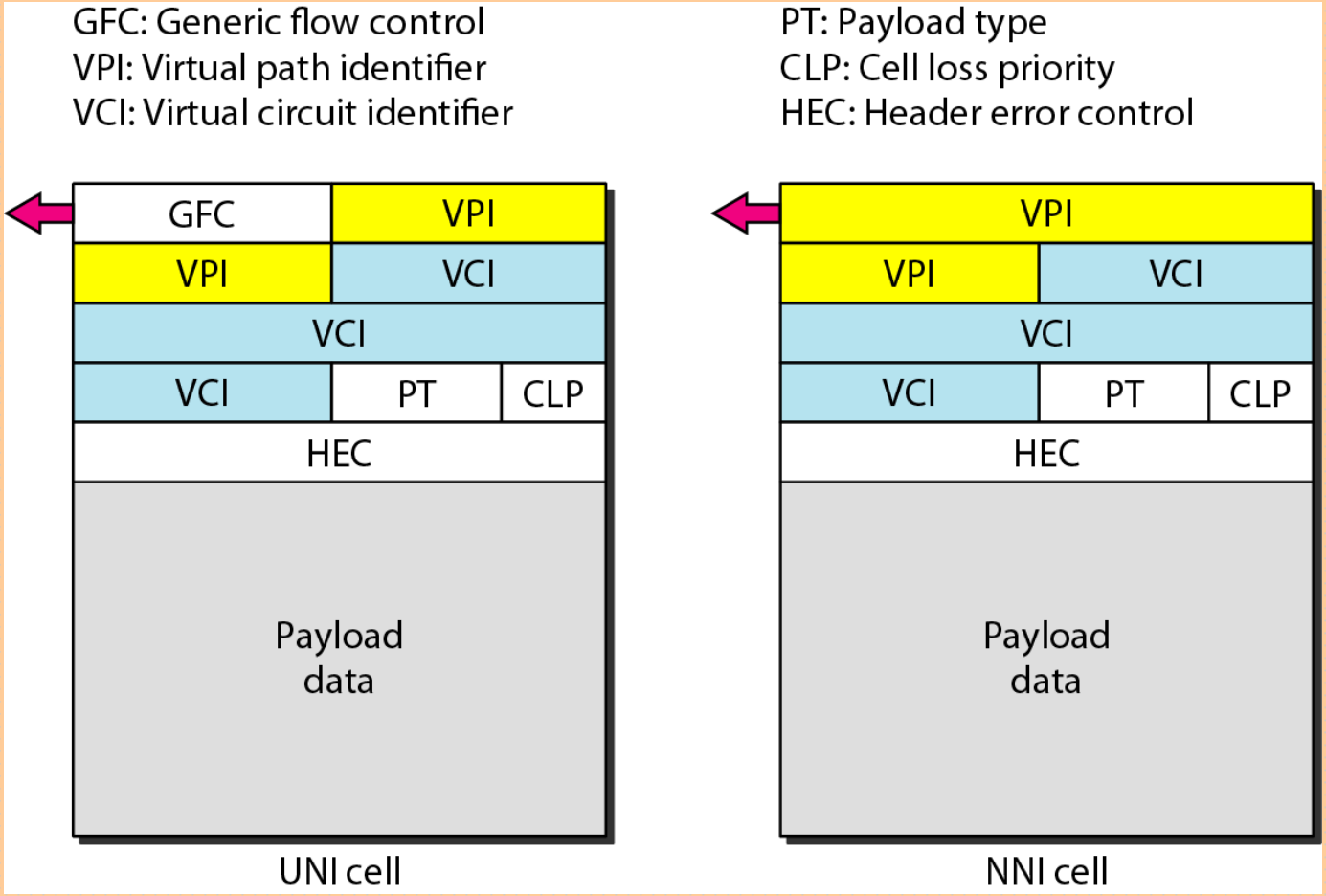
An ATM cell

The basic data unit in an ATM network is called a cell.



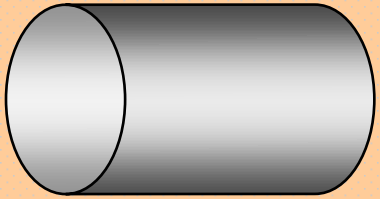
Structure of an ATM Cell

ATM headers



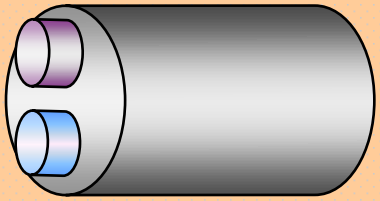
Virtual Circuits

First we have the cable...



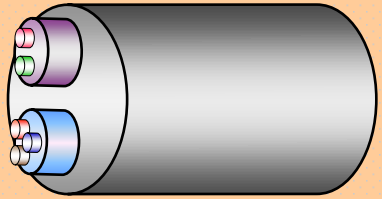
Next, ATM Addressing Defines Paths...

- **VP's**

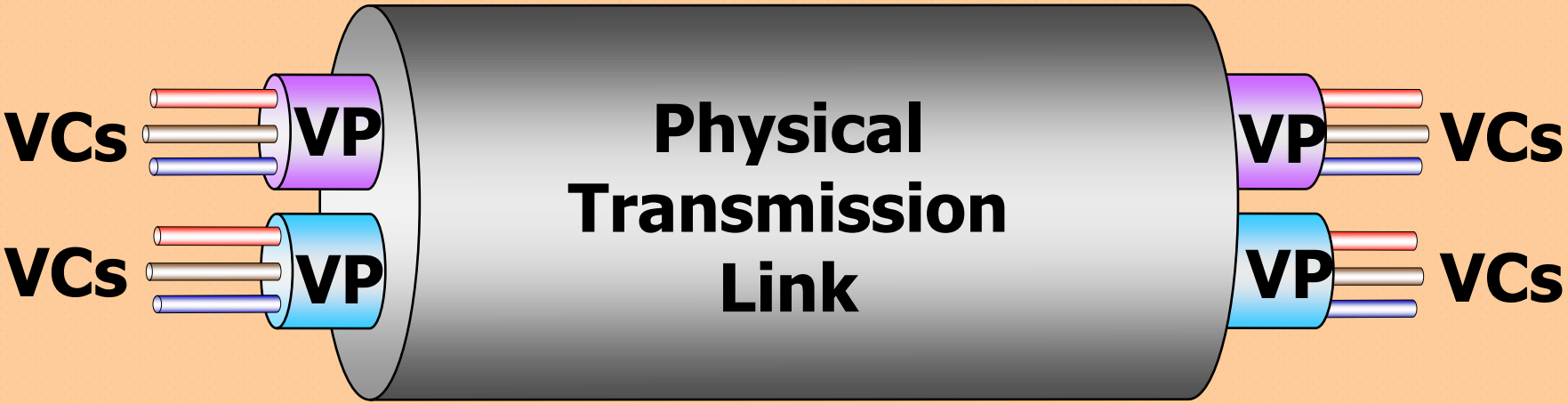


Then Channels.

- **VC's**

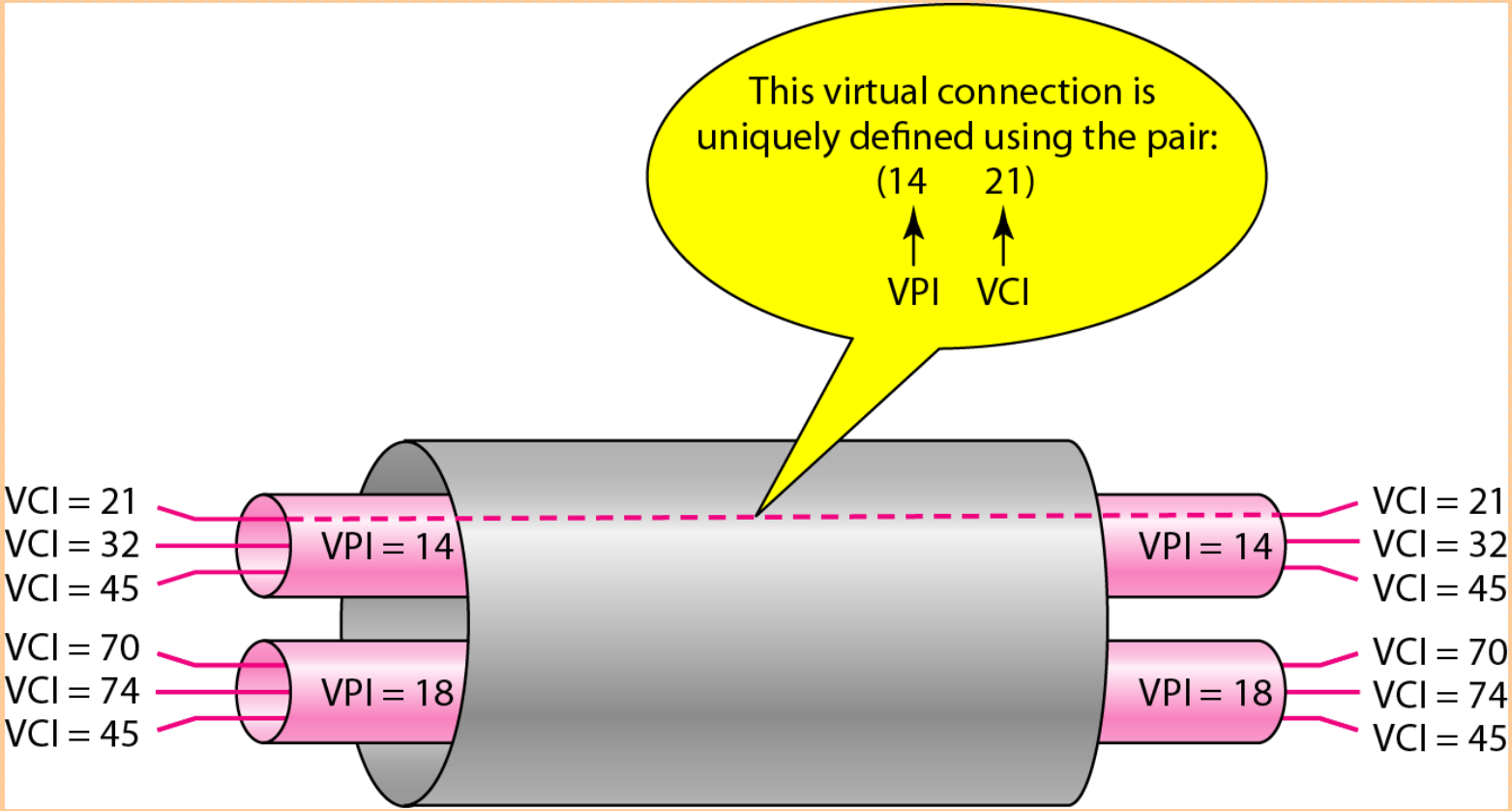


Virtual Paths & Virtual Channels



- **TP: Transmission link**
- **VPI: Virtual Path Identifier**
  - **4,096 at NNI and 256 at UNI**
- **VCI: Virtual Channel Identifier**
  - **65,536**
- **Both used to route cells through network**
  - **Unique on link-by-link basis**
  - **Interpreted at each switch**

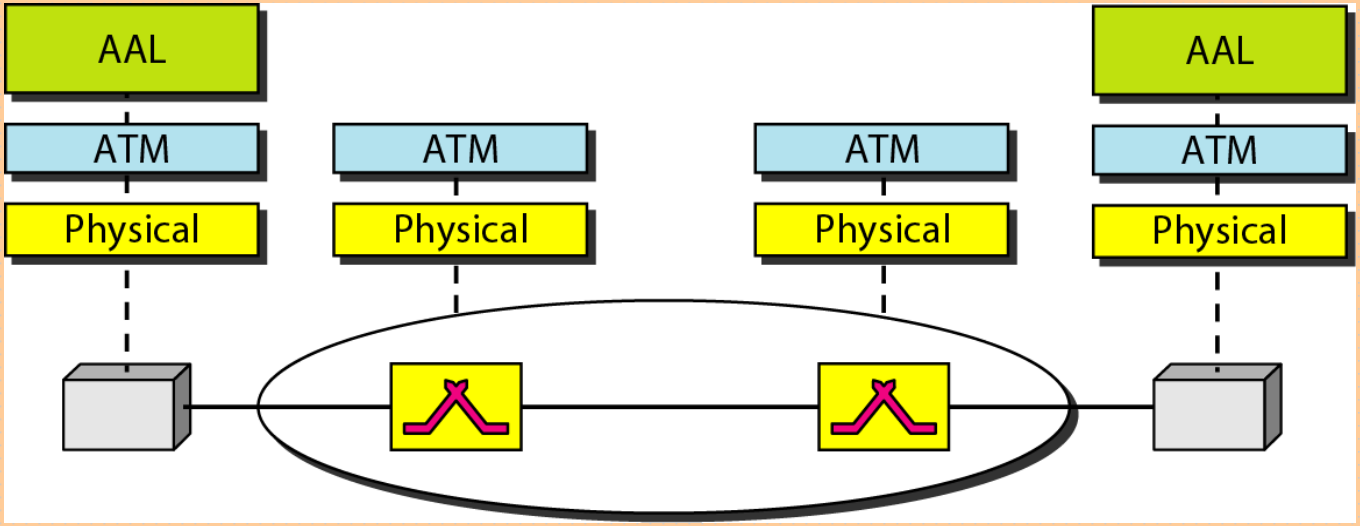
Concept – VPs and VCs in the Network



Routing Concept in an ATM Network

ATM Protocol Stack

ATM layers in endpoint devices and switches



The original design of ATM was based on SONET (Synchronous Optical Network): as the physical layer carrier:

- First, the high data rate of SONET
- Second, in SONET, the boundaries of cells can be clearly defined.

Other Physical Technologies : ATM does not limit the physical layer to SONET. Other technologies such as wireless may be used.

ATM Switches are easily Scaleable in Speed

- **ATM protocol is connection-oriented**
  - **Once connection is set up, cells are quickly switched in hardware by using VPI/VCI at very high speeds.**
- **Uses fixed cell length**
  - **Allows switch hardware to be optimized around a fixed length cell.**
- **Uses SONET as physical layer interface**
  - **Scales to high speed and is defined and deployed at Gigabit rates.**

Concept - ATM Service Categories

- **Constant Bit Rate (CBR)**
  - Used for applications requiring a fixed, guaranteed bandwidth, such as voice calls and video conferencing. (e.g., T-carrier, E-carrier)
- **Realtime Variable Bit Rate (rt-VBR)**
  - Like CBR, but the rate can transmit varies (e.g., compressed video, voice, multimedia streaming).
- **Non-realtime Variable Bit Rate (nrt-VBR)**
  - Examples of nrt-VBR traffic include data traffic and non real-time buffered voice and video traffic.(e.g., one-way TV distribution)

Concept - ATM Service Categories

- **Available Bit Rate (ABR)**
  - For applications that can adjust their data rate based on available network capacity, such as file transfers (bursty LAN traffic) and e-mail.

- **Unspecified Bit Rate (UBR)**
  - Used for best-effort traffic like email and file downloads that do not require strict QoS.

Revised ATM Service Categories

Class	Description	Example
CBR	Constant Bit Rate	T-carrier, E-carrier
RT-VBR	Real Time Variable Bit Rate	Real-time videoconferencing
NRT-VBR	Non-real-time Variable Bit Rate	Multimedia email
ABR	Available Bit Rate	Browsing the Web
UBR	Unspecified Bit Rate	Internet access file download

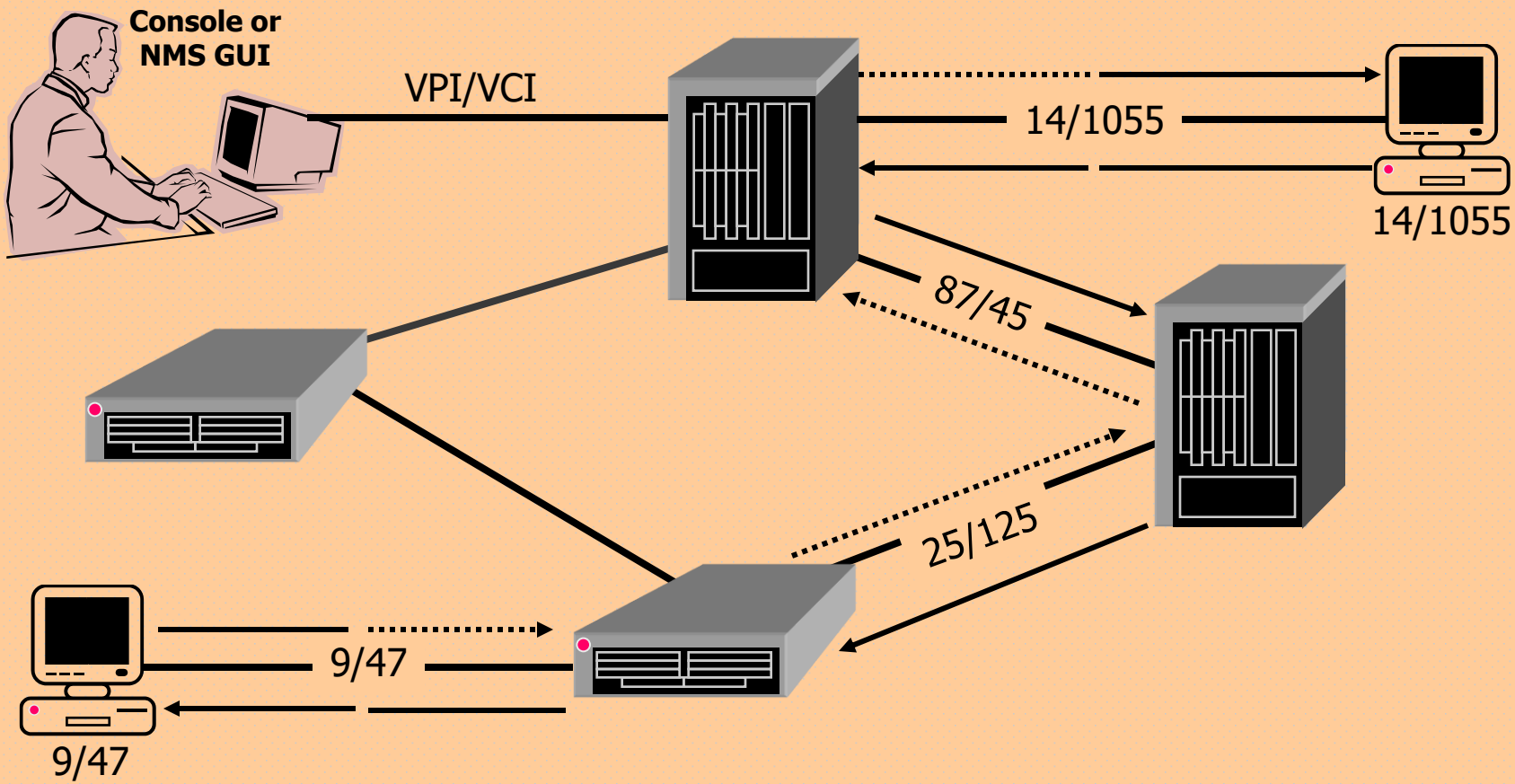
Connection Establishment and Release

- ATM uses two types of connections : PVC and SVC

- PVC : a permanent virtual-circuit is established between two endpoints by the network provider. The VPIs and VCIs are defined for the permanent connection and the values are entered for the tables of each switch.

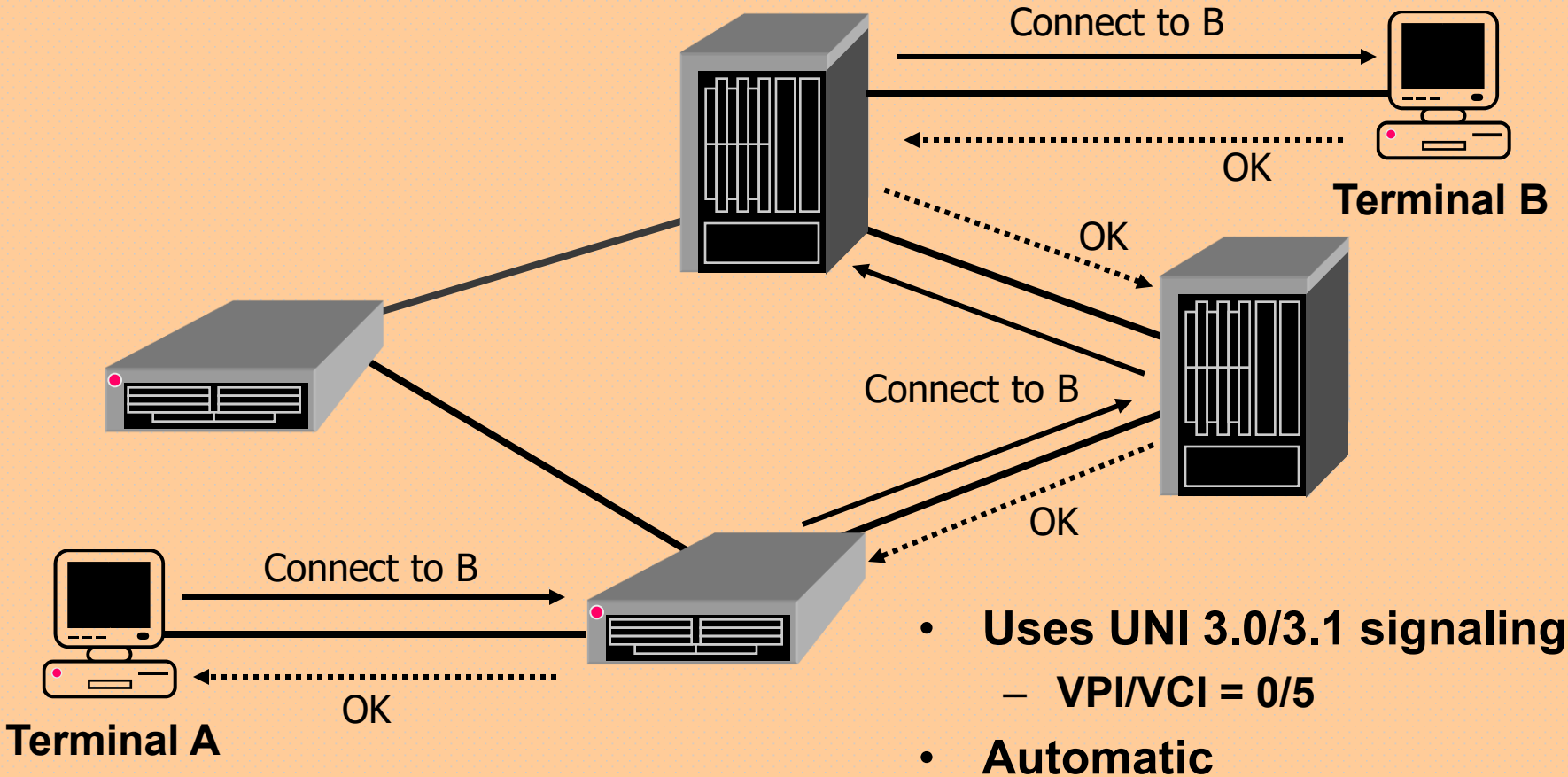
- SVC : In a switched virtual-circuit connection, each time an endpoint wants to make a connection with another endpoint, a new virtual circuit must be established. ATM cannot do the job by itself, but needs the network layer addresses and the services of another protocol (such as IP).

PVC - Manual Set Up



- **Pre-established connections**
- **Permanent**
- **No signaling required**

SVC - Automatic Set Up



- **Uses UNI 3.0/3.1 signaling**
  - **VPI/VCI = 0/5**
- **Automatic**
- **Transparent to User**

UNI 3.0/3.1 signaling with VPI/VCI 0/5 is standard signaling protocol for setting up connections in an ATM networks.

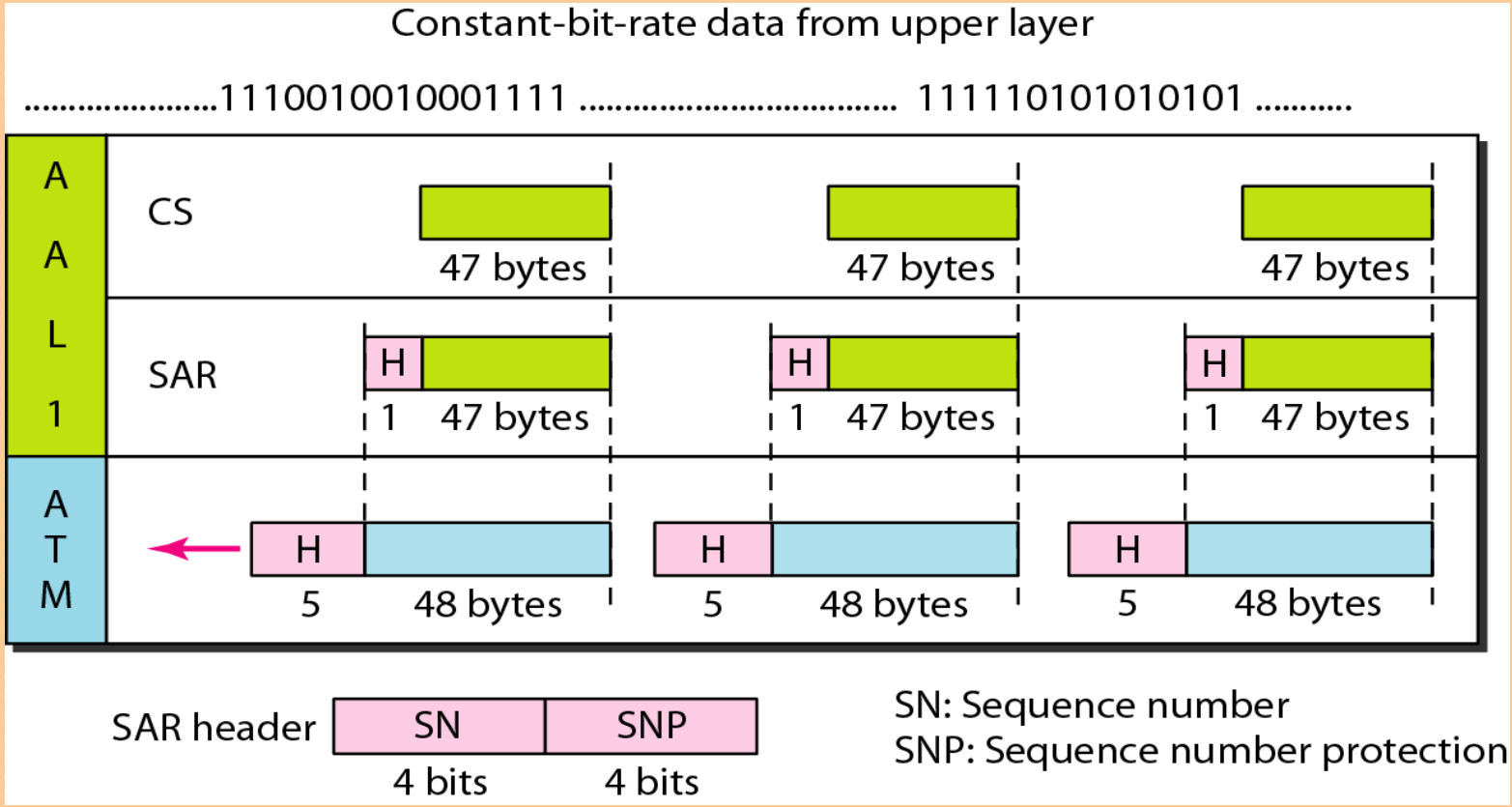
Application Adaptation Layer (AAL)

- **The AAL defines a sublayer, called a segmentation and reassembly (SAR) sublayer, and called the convergence sublayer (CS).**

- **ATM defines four versions of the AAL: AAL1, AAL2, AAL3/4, and AAL5.**

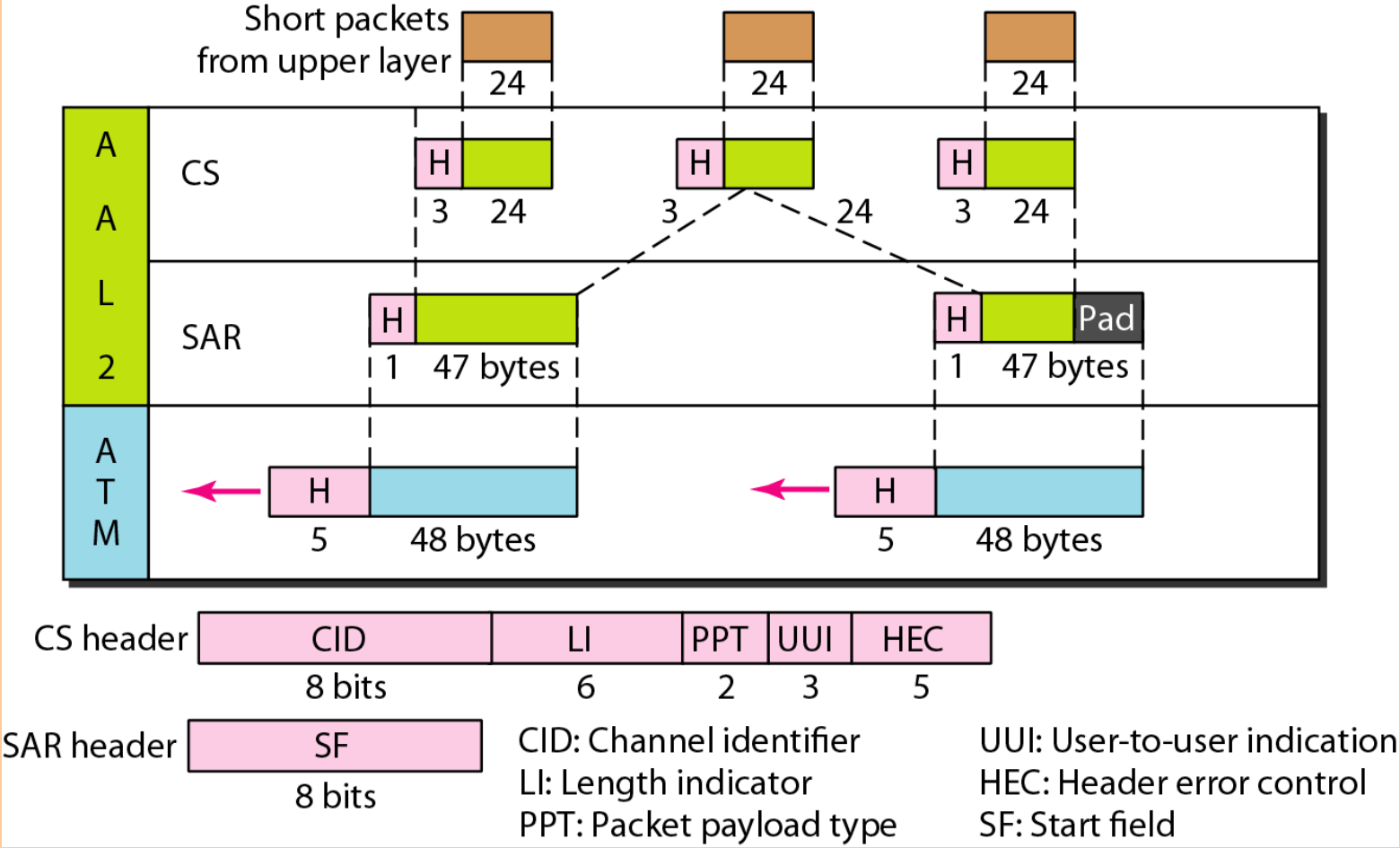
- **The common versions today are AAL1 and AAL5. The first is used in streaming audio and video communication; the second, in data communications.**

AAL1



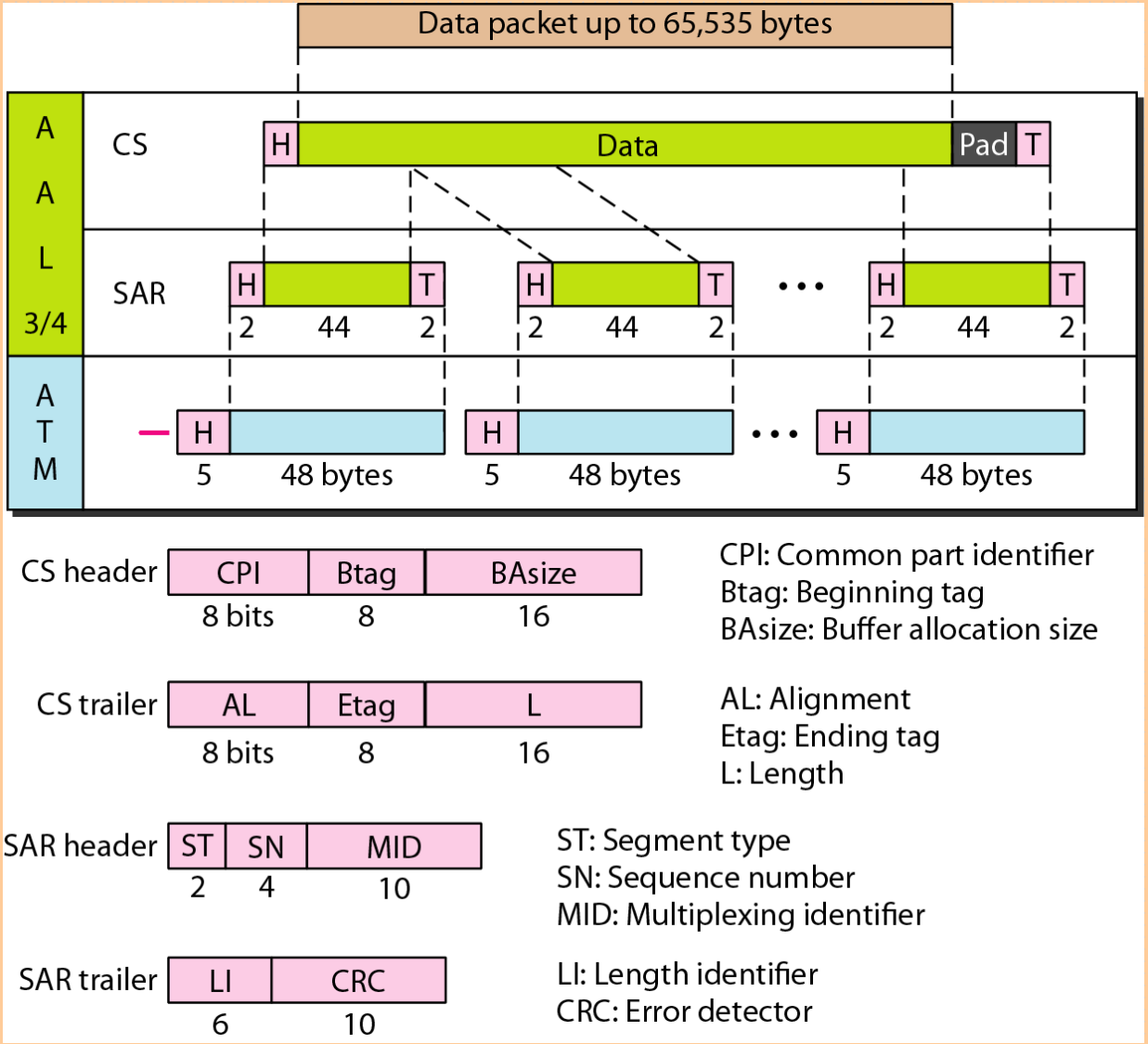
Supports applications that transfer information at constant bit rates such as video and voice. It allows ATM to connect existing digital telephone networks such as voice channels and T lines.

AAL2



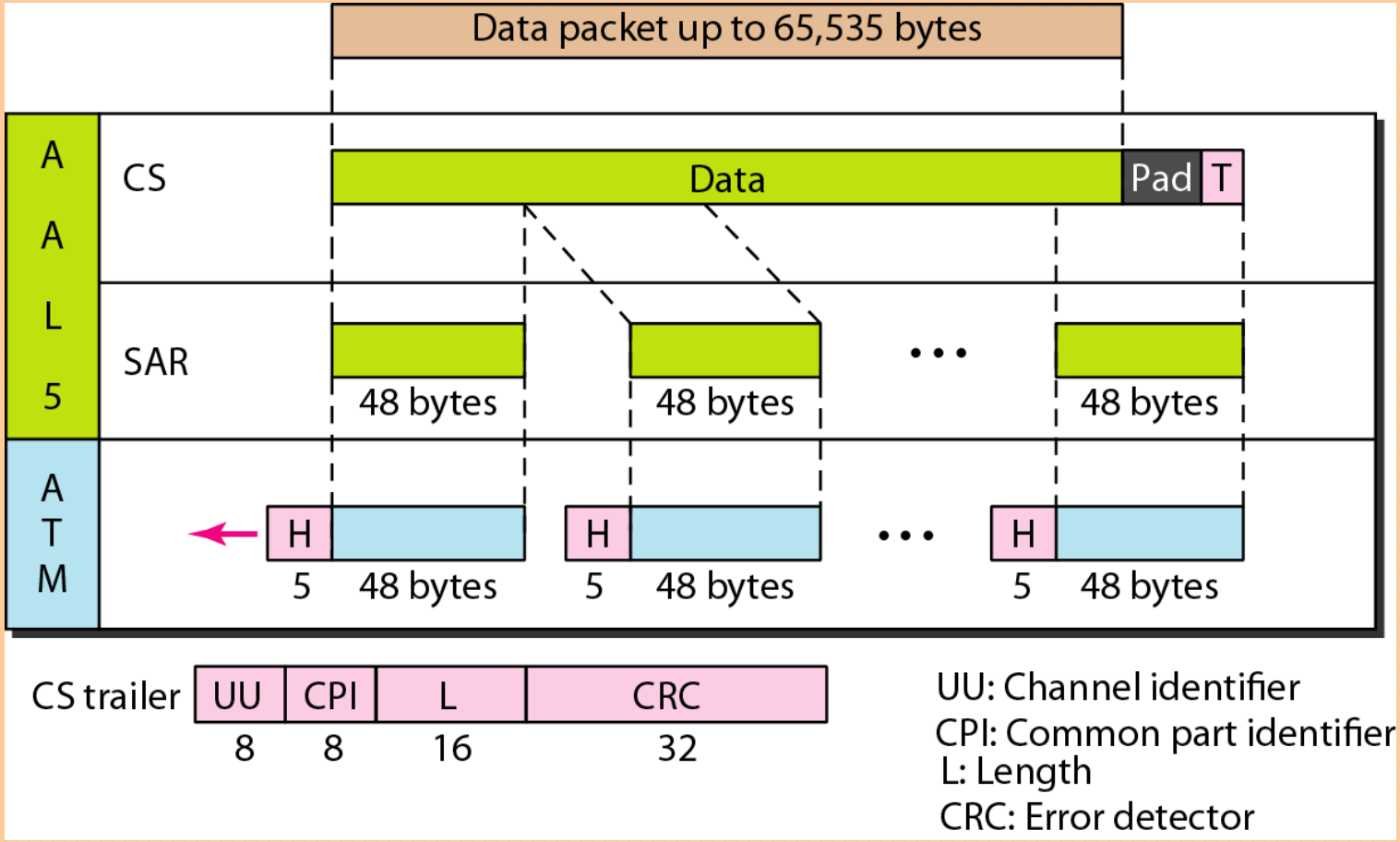
It is used for low bit rate traffic and short-frame traffic such as audio, video.
Example of AAL2 use is in mobile telephony.

AAL3/4



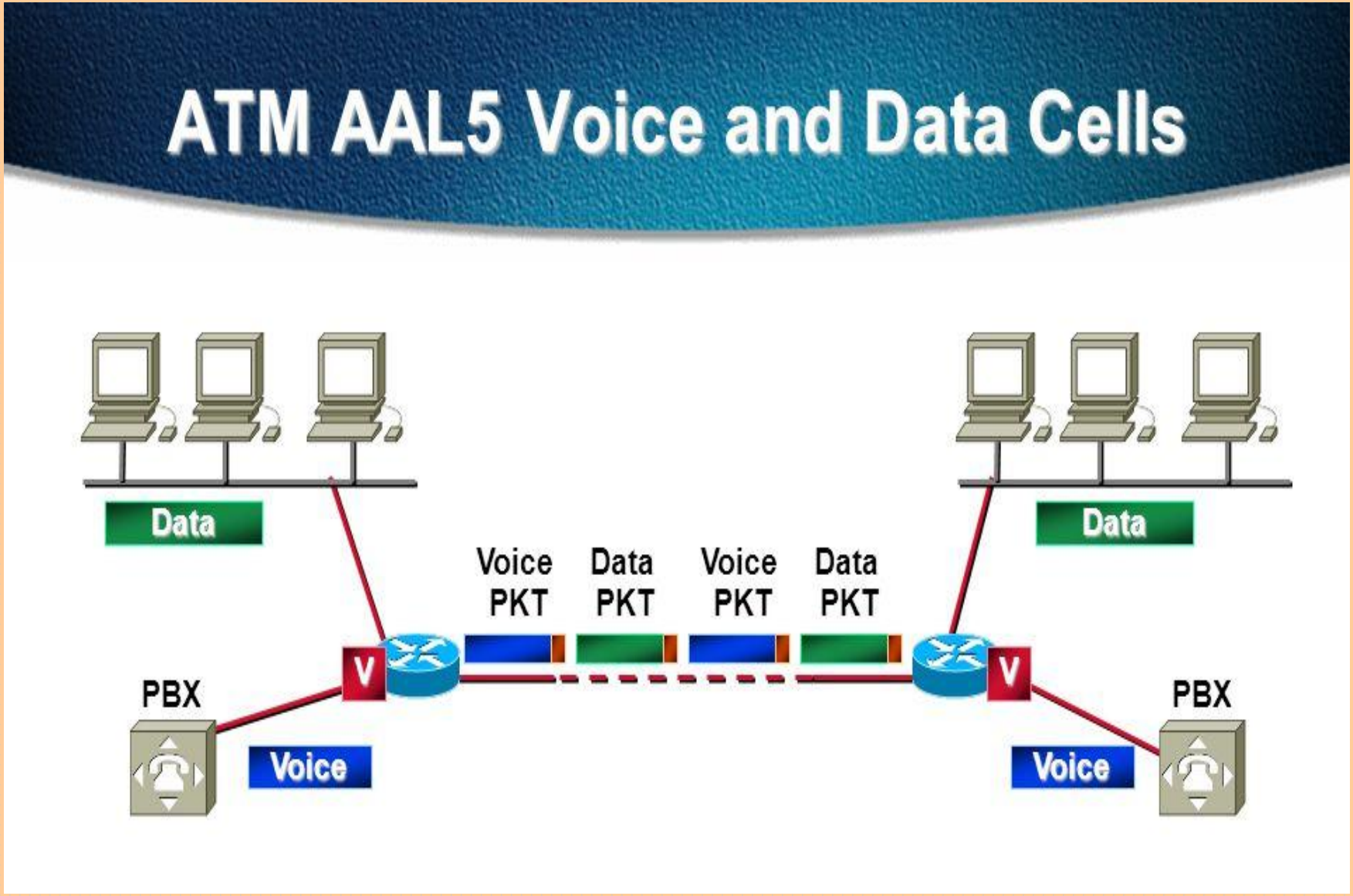
AAL3 was intended to support connection-oriented data services and AAL4 to support connectionless services.

AAL5



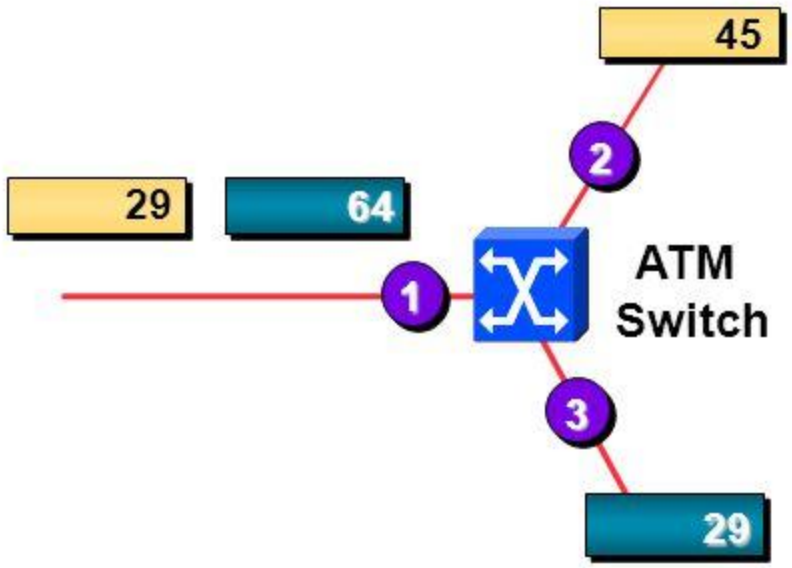
AAL5 is less complex than type 3/4 by reduce transmission overhead, and popular used in ATM LAN application.

AAL5


ATM AAL5 Voice and Data Cells
Data
Data
Voice PKT
Data PKT
Voice PKT
Data PKT
V
V
PBX
PBX
Voice
Voice

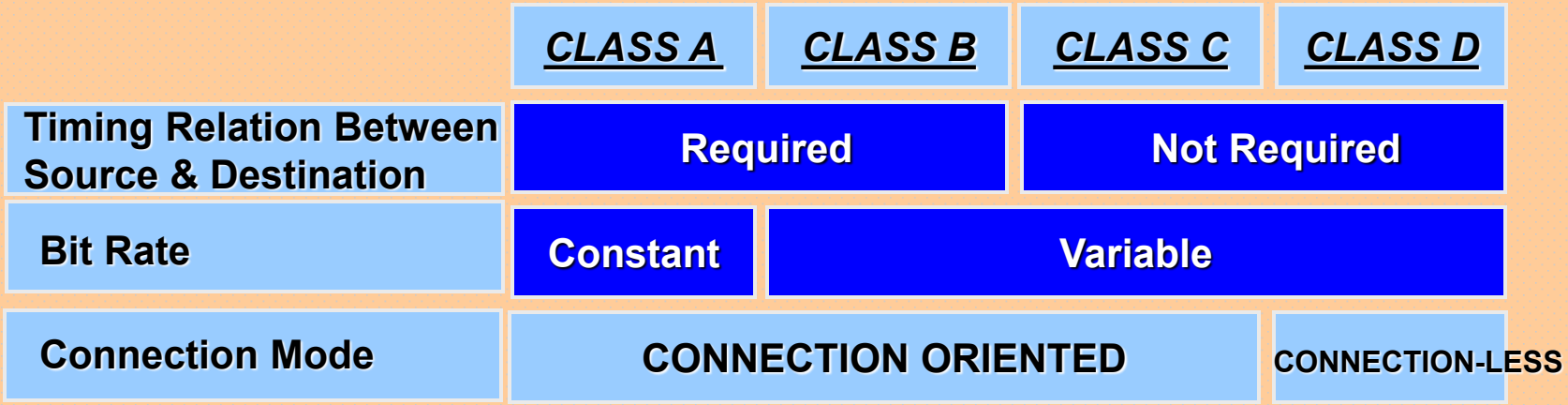
ATM Switches

Input		Output	
Port	VPI/VCI	Port	VPI/VCI
1	29	2	45
2	45	1	29
1	64	3	29
3	29	1	64



- ATM switch performs cell-relay using connection identifiers
  - Virtual Channel Identifier (VCI)
  - Virtual Path Identifier (VPI)
- ATM layer in ATM switches translates VPI/VCI values
- VPI/VCI value unique only per interface, may be re-used elsewhere in network

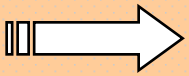
Classes of ATM Service (ITU)



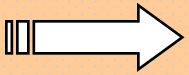
ATM

- ATM is a new switching and multiplexing technique designed for broadband, multi-service communication networks.

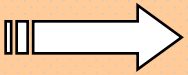
จุดเด่น คือ รองรับข้อมูลหลายรูปแบบ รับประกันคุณภาพบริการ และการส่งข้อมูลในรูปแบบเซลล์ขนาดคงที่ Low Latency and Minimal Jitter



VOICE

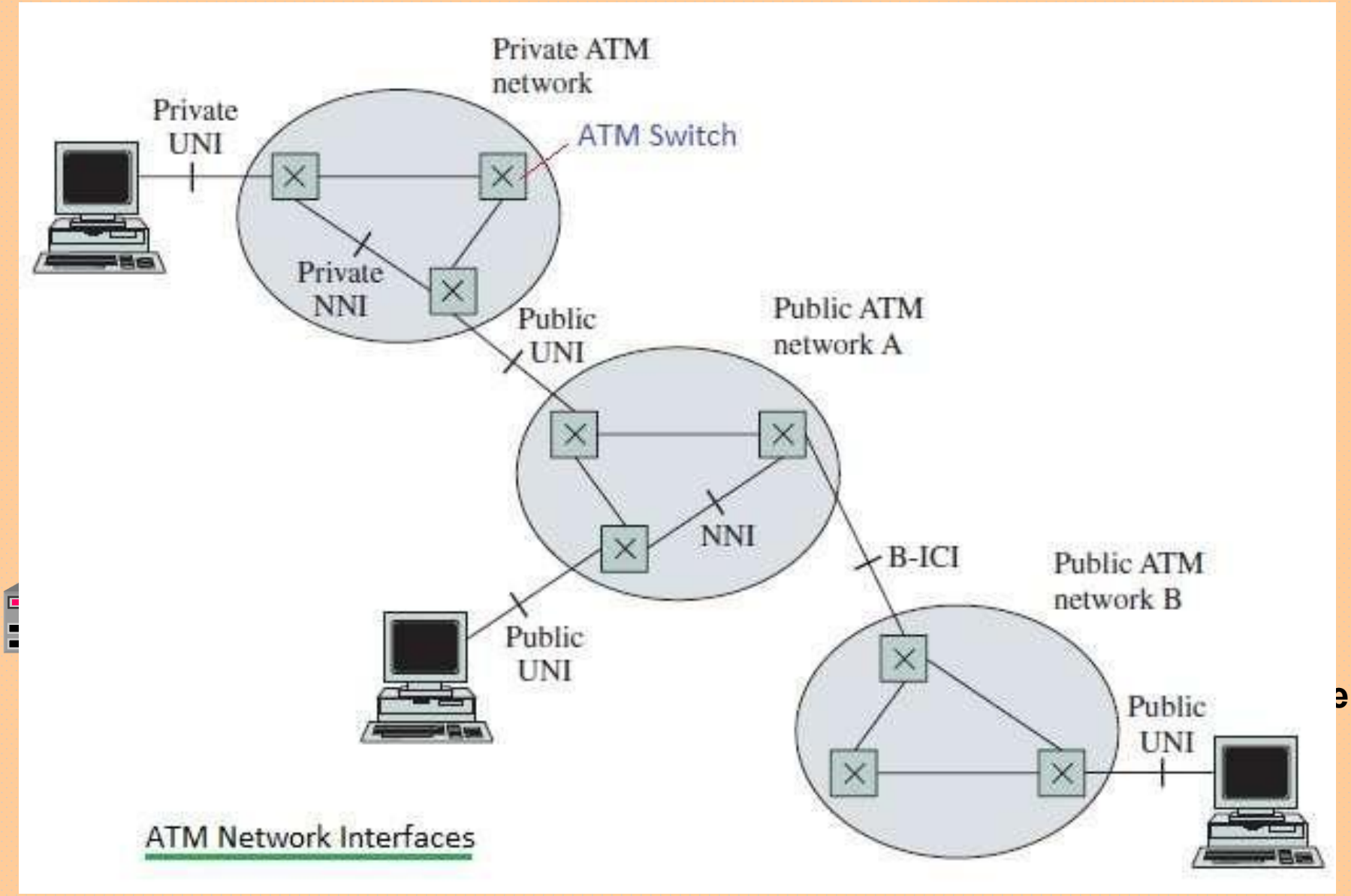


DATA

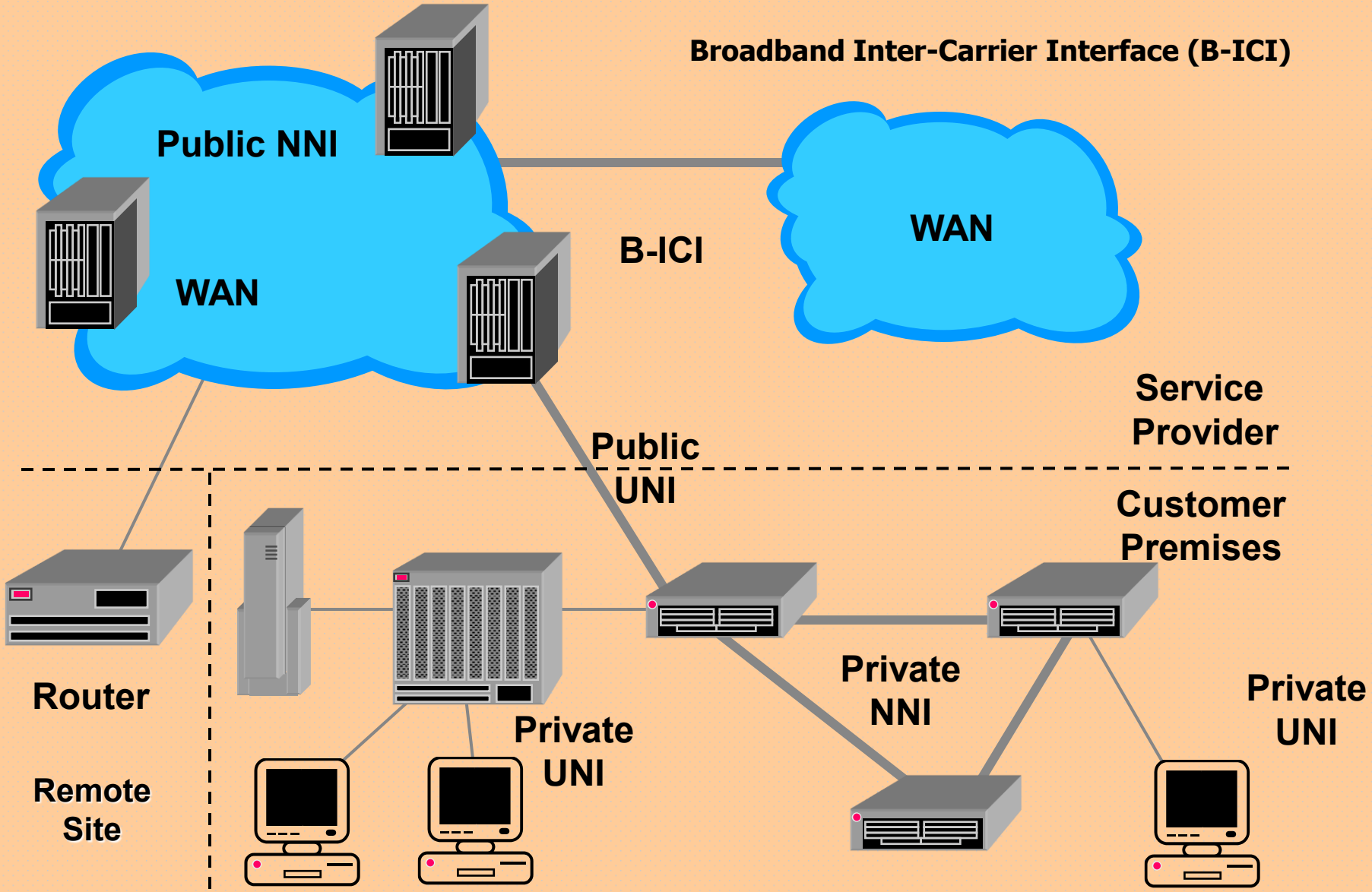


VIDEO

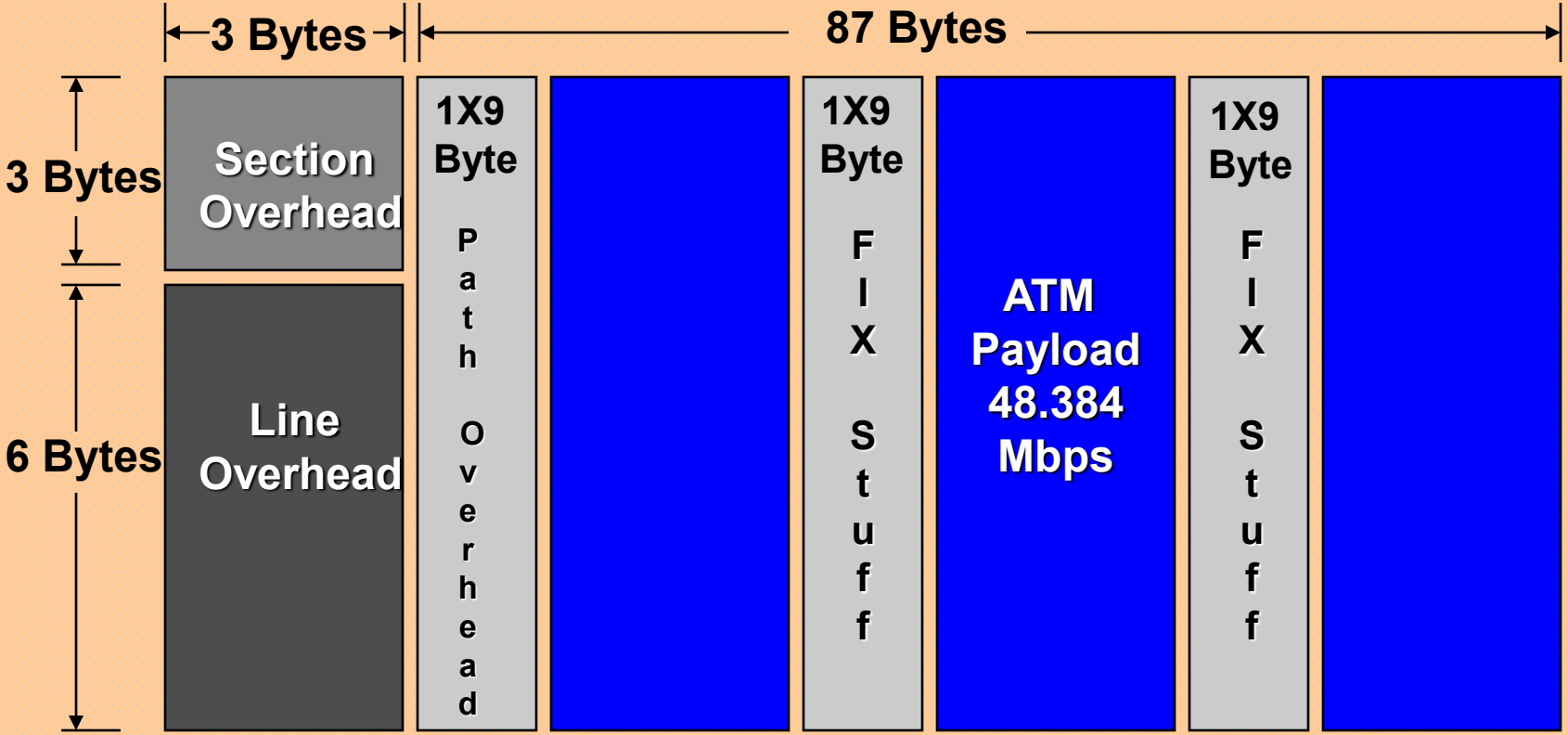
ATM Network Interfaces



ATM Network Interfaces

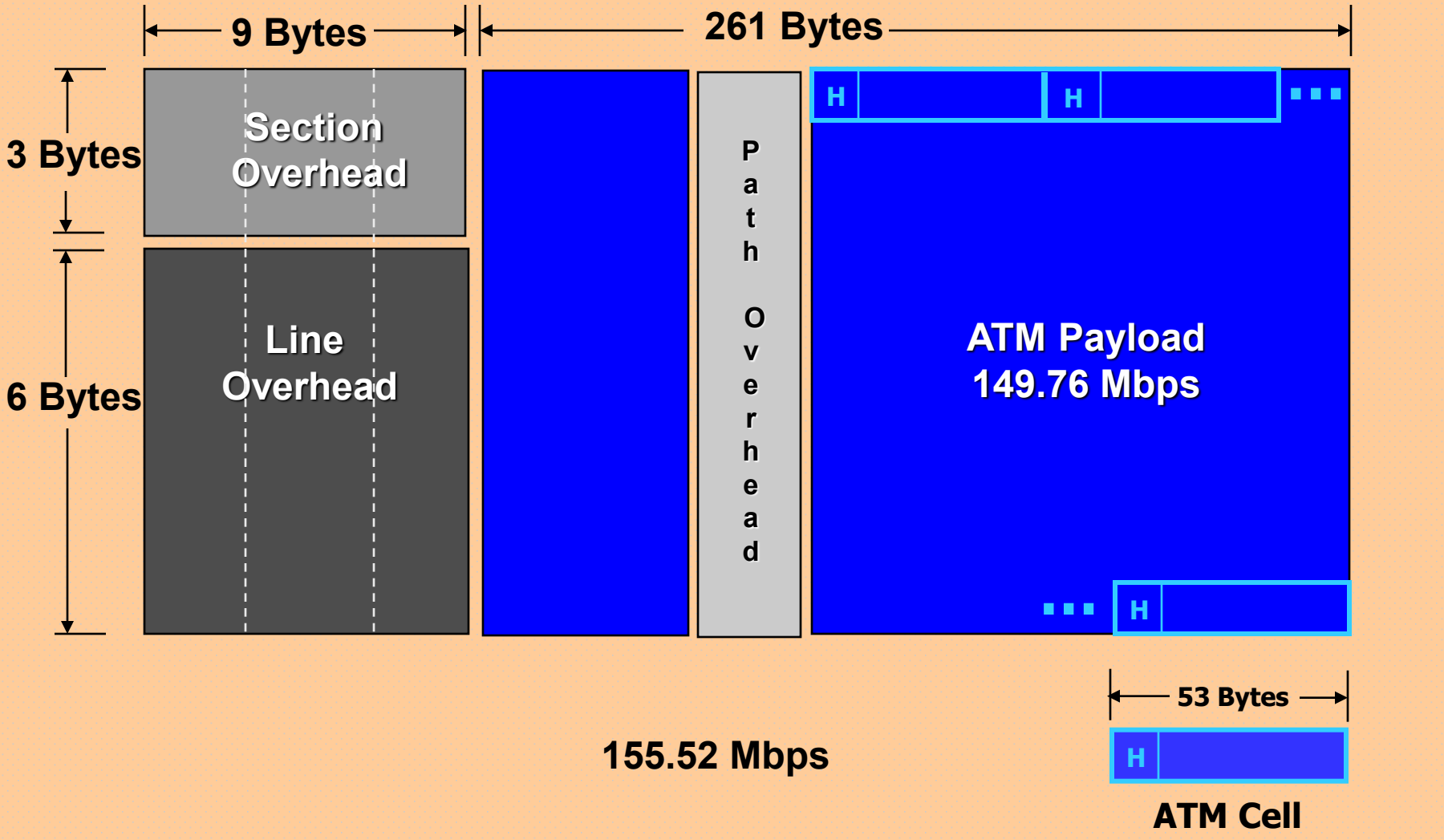


STS- 1 Frame with ATM

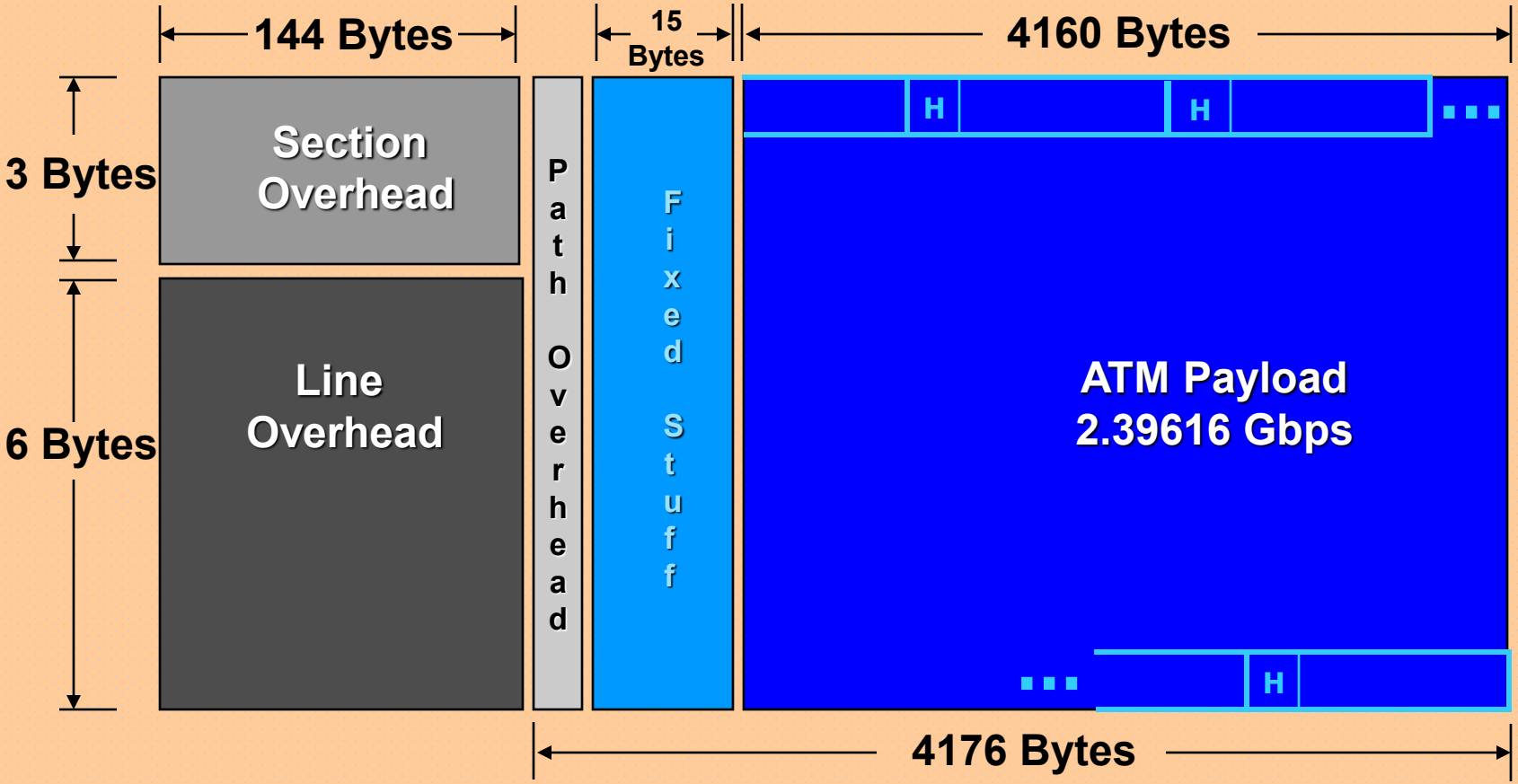


51.84 Mbps

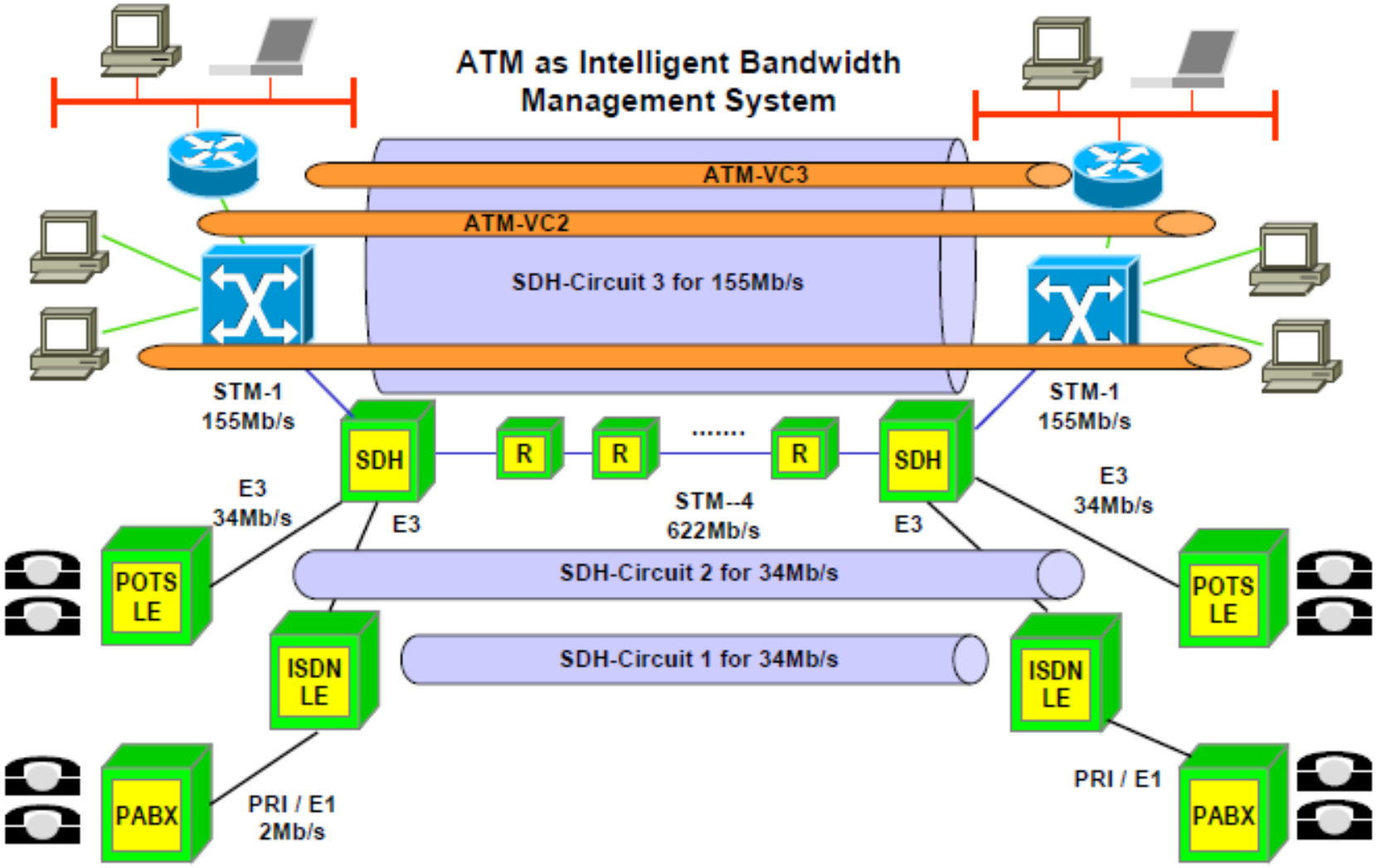
STS-3c Frame Structure



STS-48c Frame Structure



ATM-VCs inside SDH-Circuit

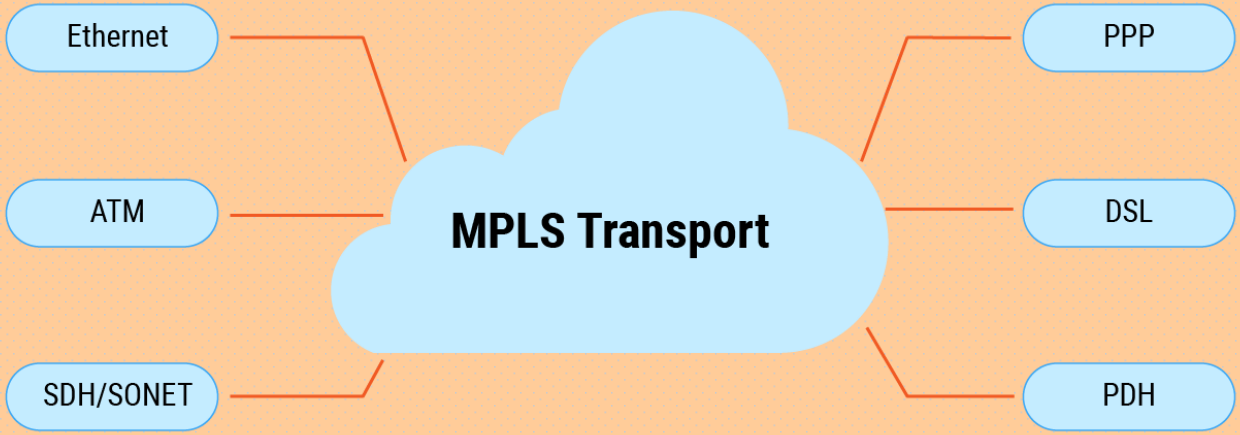


REFERENCES

- *ATM Forum Documents - www.atmforum.com*
- *GR-1248 - Generic Requirements for Operations on ATM NEs*
- *I.363 & 365 - ATM Adaptation Layer (AAL) Sublayer*

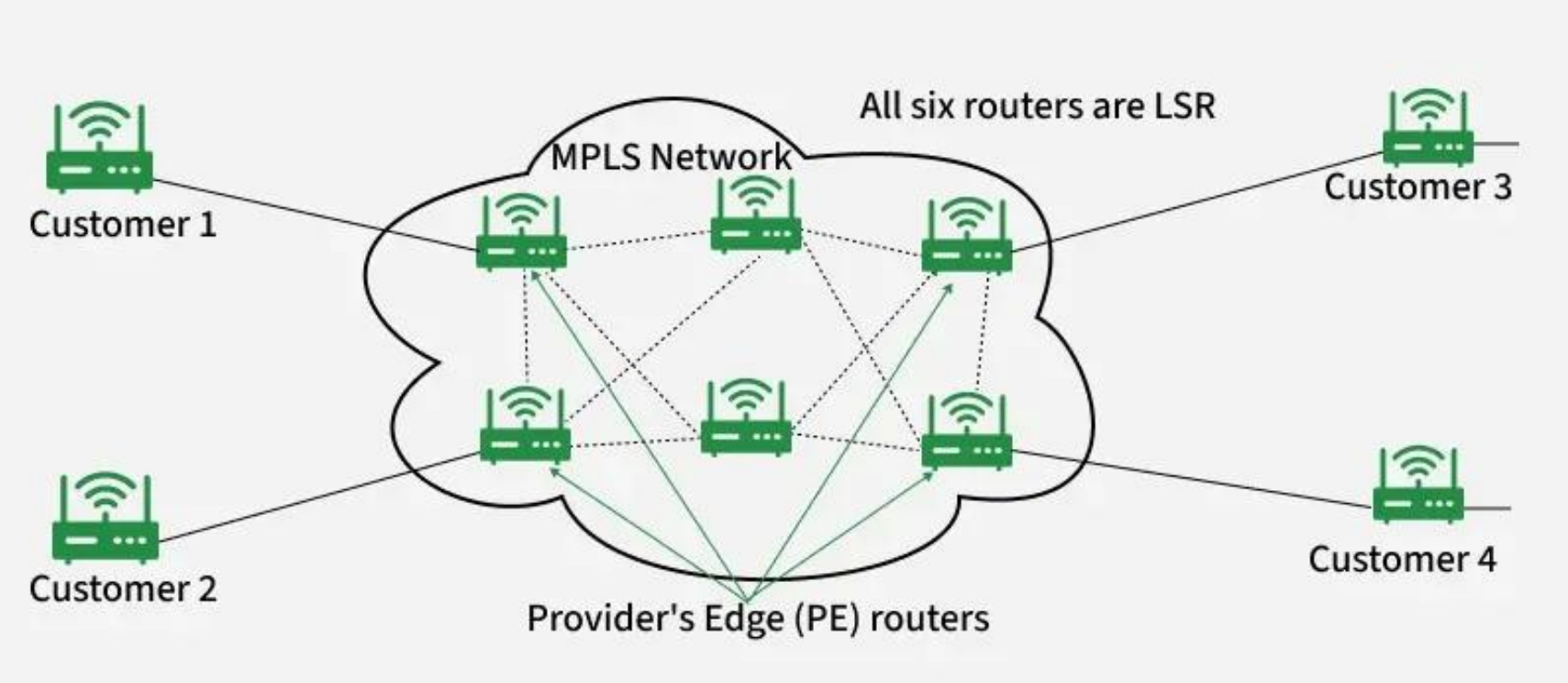


Multi-Protocol Label Switching (MPLS)

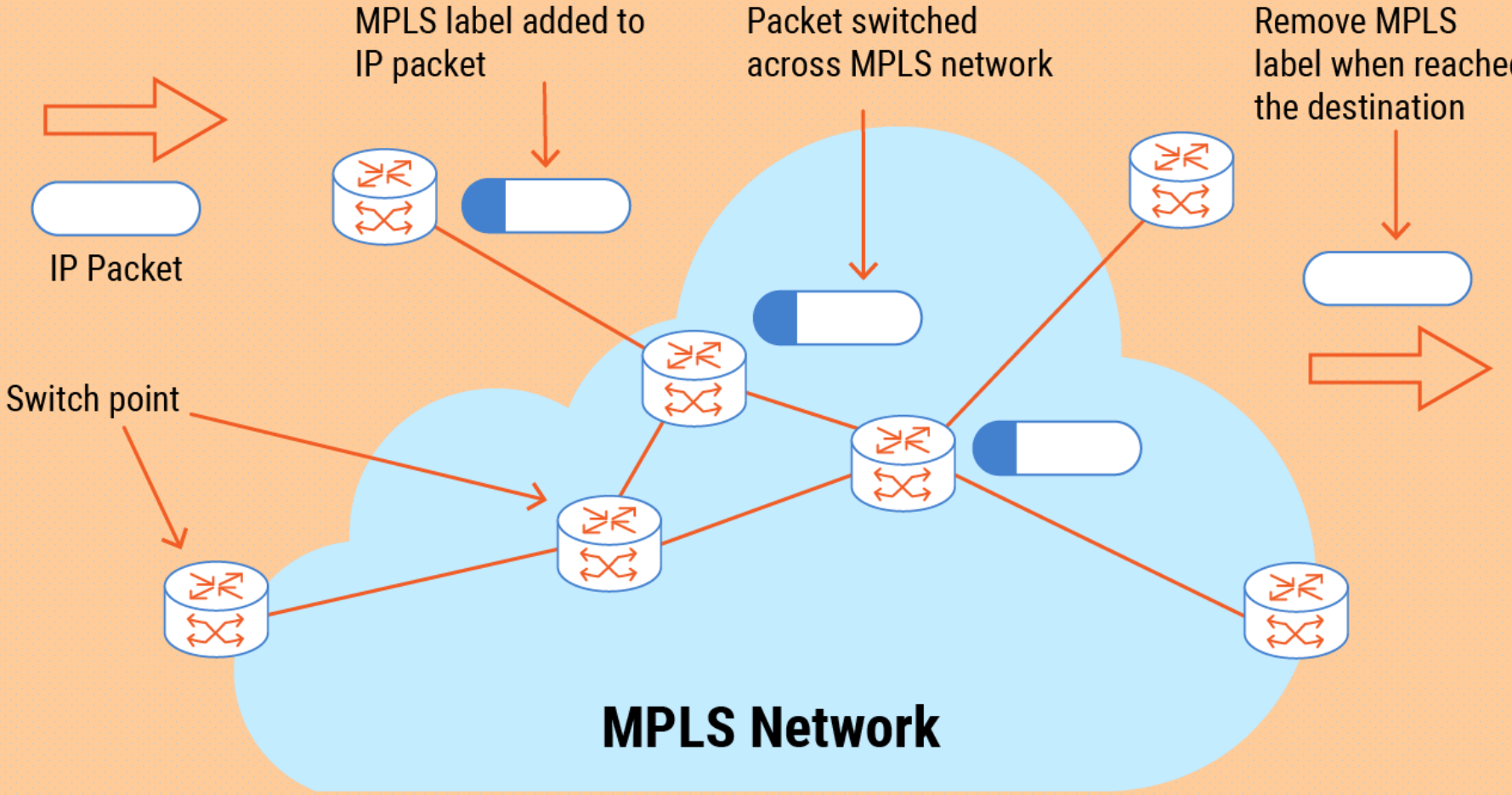


https://www.cisco.com/c/dam/global/fr_ca/training-events/pdfs/Intro_to_mpls.pdf
https://www.geeksforgeeks.org/computer-networks/multi-protocol-label-switching-mpls/
https://www.rfwireless-world.com/terminology/mpls-label-switching-vs-ip-packet-switching

Multi-Protocol Label Switching (MPLS) is an advanced packet-forwarding technique used in modern networks. Instead of making routers look into complex Layer 3 routing tables for every IP packet, MPLS uses labels for forwarding decisions. These labels create pre-defined, efficient paths across the network, which enhances speed, scalability and traffic management.



Note: *MPLS lies between Layer 2 (Data Link) and Layer 3 (Network Layer) of the OSI model, which is why it is often referred to as a Layer 2.5 protocol.*



https://www.iptp.net/blog/what-is-mpls-multiprotocol-label-switching/

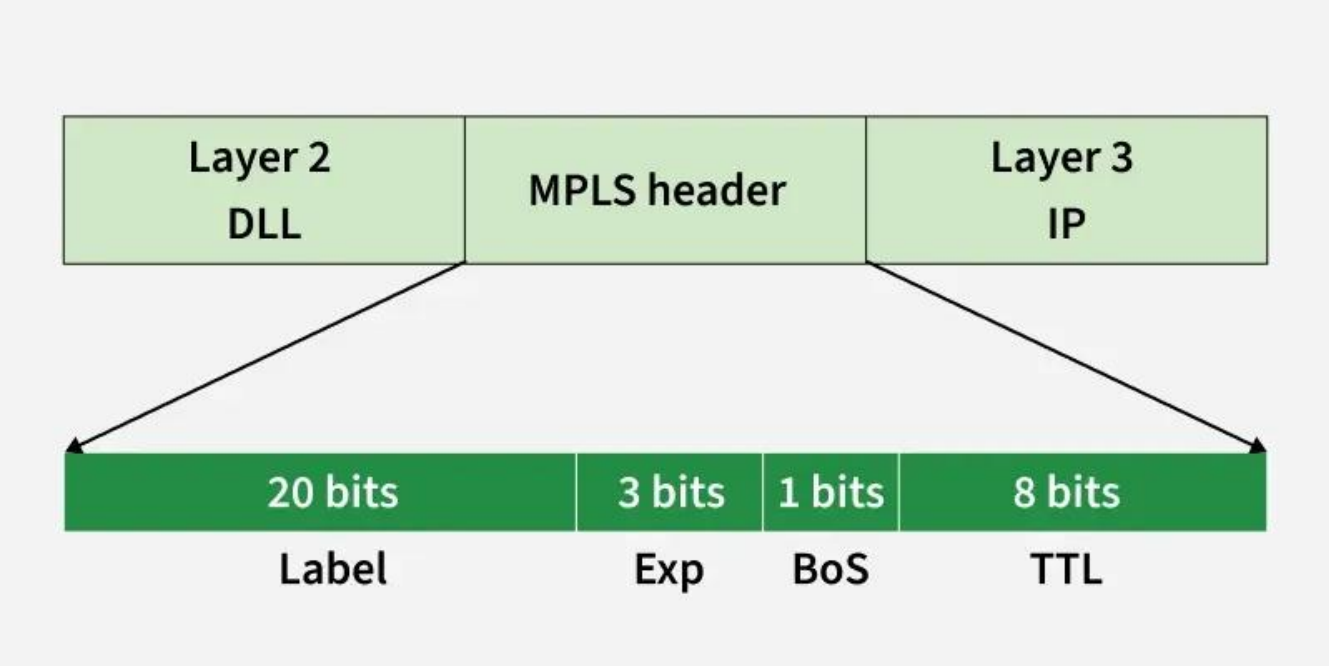
Why MPLS

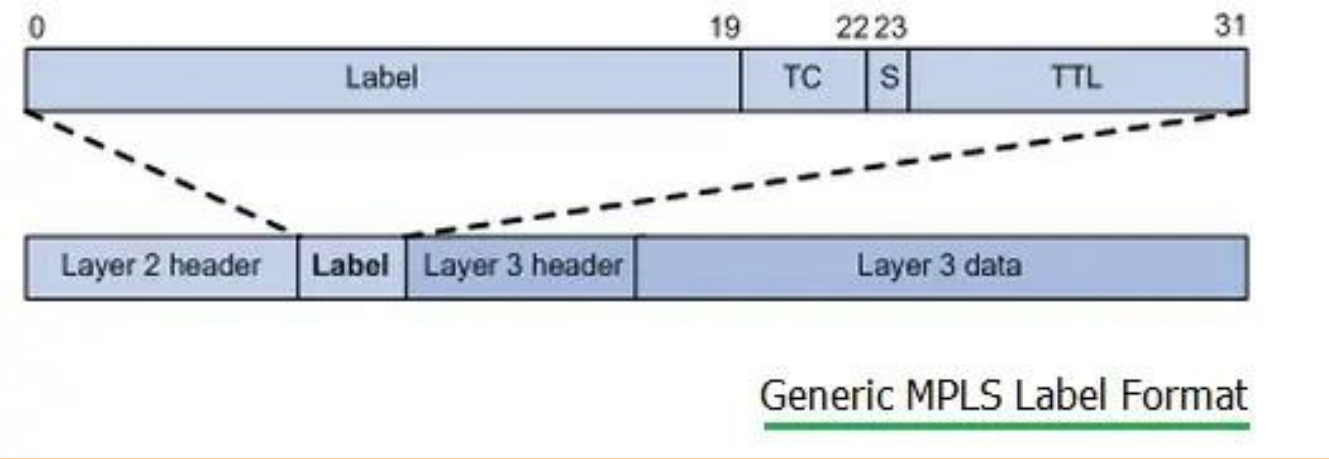
- Traditional IP routing involves a long lookup process in routing tables.
- MPLS simplifies forwarding by assigning a short fixed-length label to each packet.
- This ensures: Faster packet delivery, better support for Quality of Service (QoS) & easier traffic engineering and Virtual Private Networks (VPNs).

MPLS Header

- The MPLS header is 32 bits long and is inserted between the Layer 2 and Layer 3 headers. It contains the following fields:

MPLS Header


Layer 2
DLL
MPLS header
Layer 3
IP
20 bits
3 bits
1 bits
8 bits
Label
Exp
BoS
TTL


0
19
22 23
31
Label
TC
S
TTL
Layer 2 header
Label
Layer 3 header
Layer 3 data
Generic MPLS Label Format

An MPLS label, used for label switching, is inserted between the Layer 2 header and the Layer 3 header. It is 4 octets (32 bits) in size and consists of the following fields:

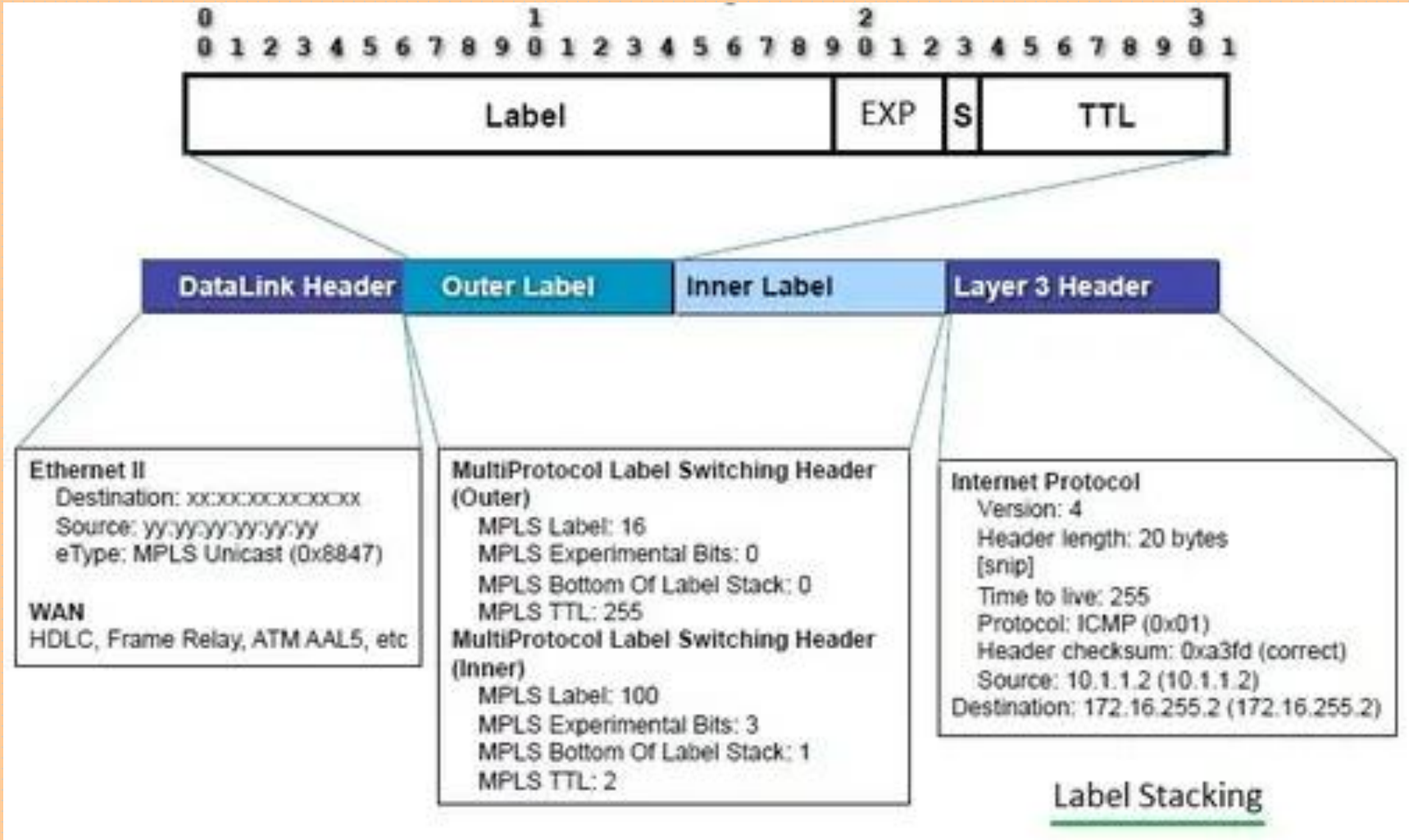
MPLS Label = { Label (20 bits) , EXP or TC (3 bits), S (1 bit), TTL (8 bits) }

- **Label (20 bits):** The actual label used for switching. Values from 0 to 15 are reserved.
- **EXP (3 bits):** Experimental bits, often used to indicate Quality of Service (QoS) or priority for the label. Also known as Traffic Class (TC).
- **S (1 bit):** Indicates whether the label is the last one in the stack. A value of 1 indicates it's the bottom label.
- **TTL (8 bits):** Time to Live.

Field	Size	Description
Label	20 bits	Identifier used to make forwarding decisions (range: 0 to $2^{20} - 1$).
Experimental (Exp)	3 bits	Used for Quality of Service (QoS) and traffic prioritization.
Bottom of Stack (S)	1 bit	Indicates whether this label is the last in the stack (1 = last label).
Time to Live (TTL)	8 bits	Prevents loops; decrements by 1 at every hop.

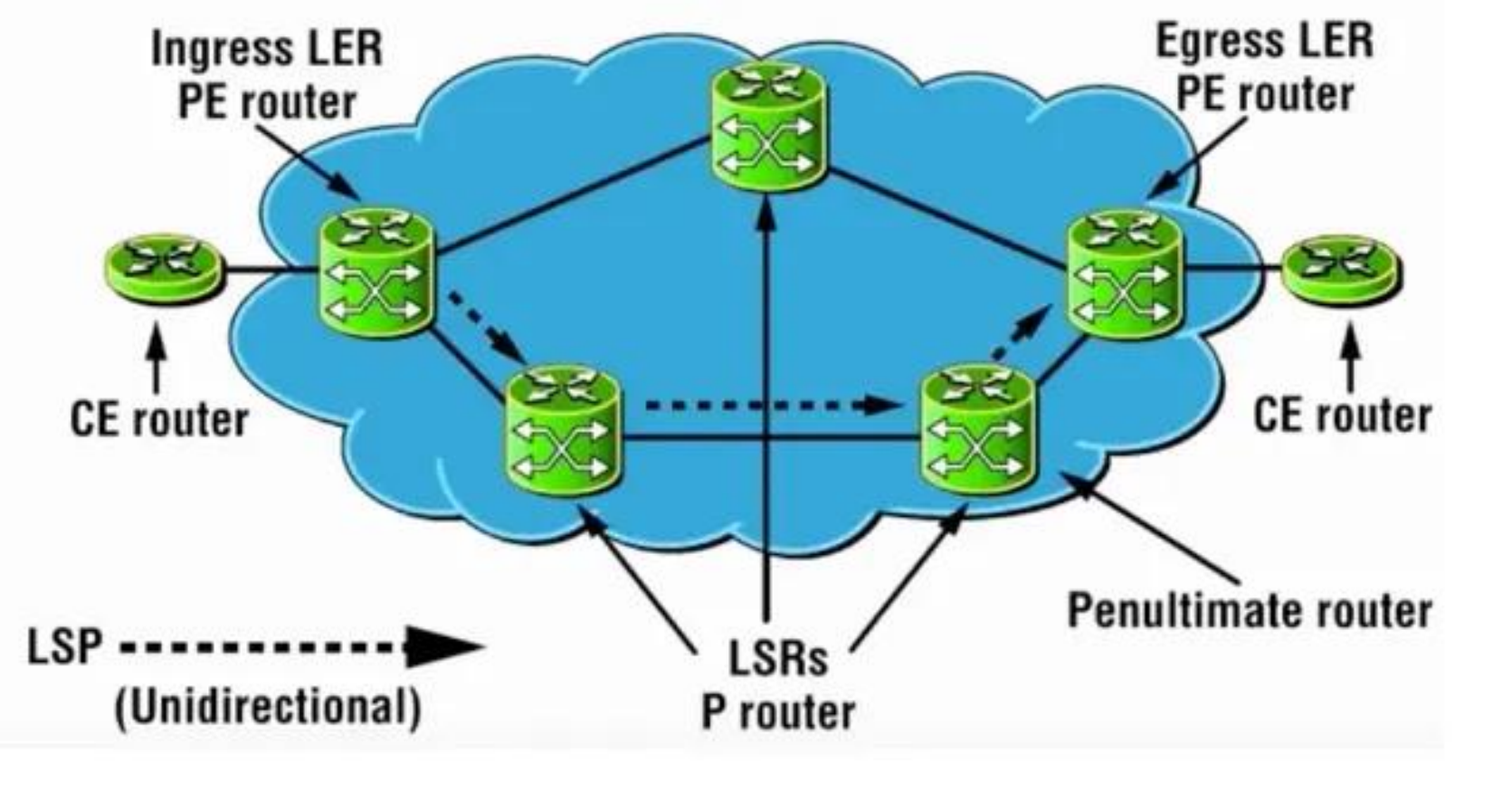
Note: *Multiple labels can be stacked to form a label stack, enabling hierarchical routing.*

Label Stacking

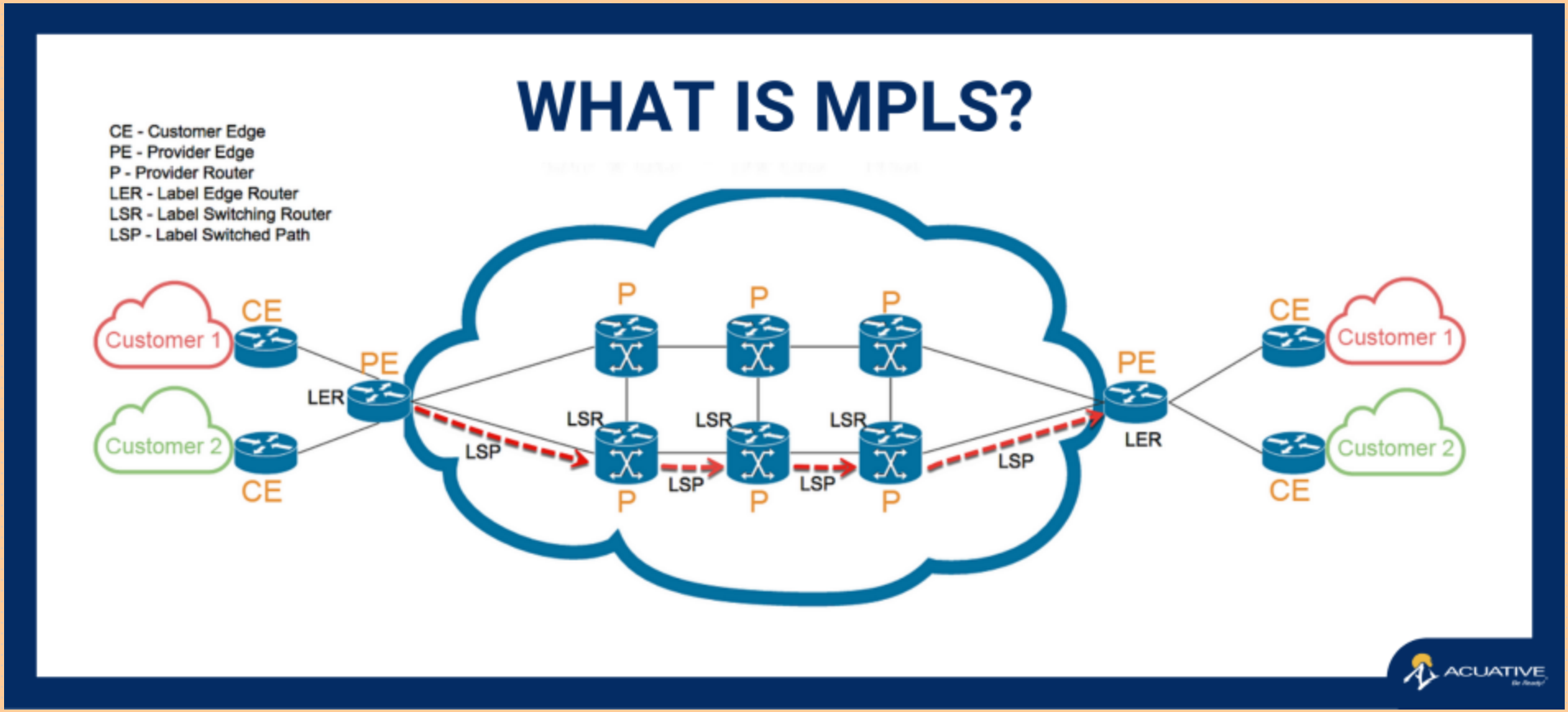


- Multiple labels can be applied to a single packet.
- The label closest to the Layer 2 header is known as the “top label or outer label”, while the label closer to the Layer 3 header is the “bottom label or inner label”.

MPLS Network Architecture



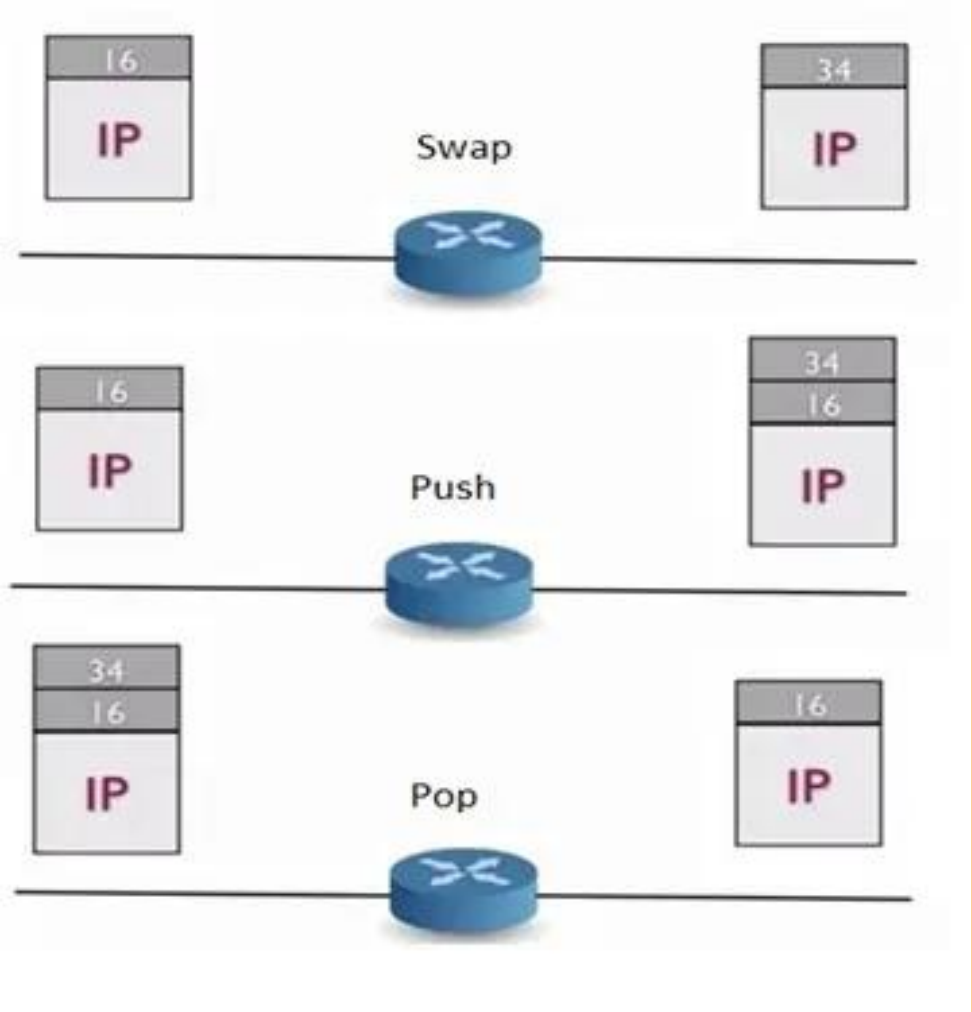
MPLS Network Architecture



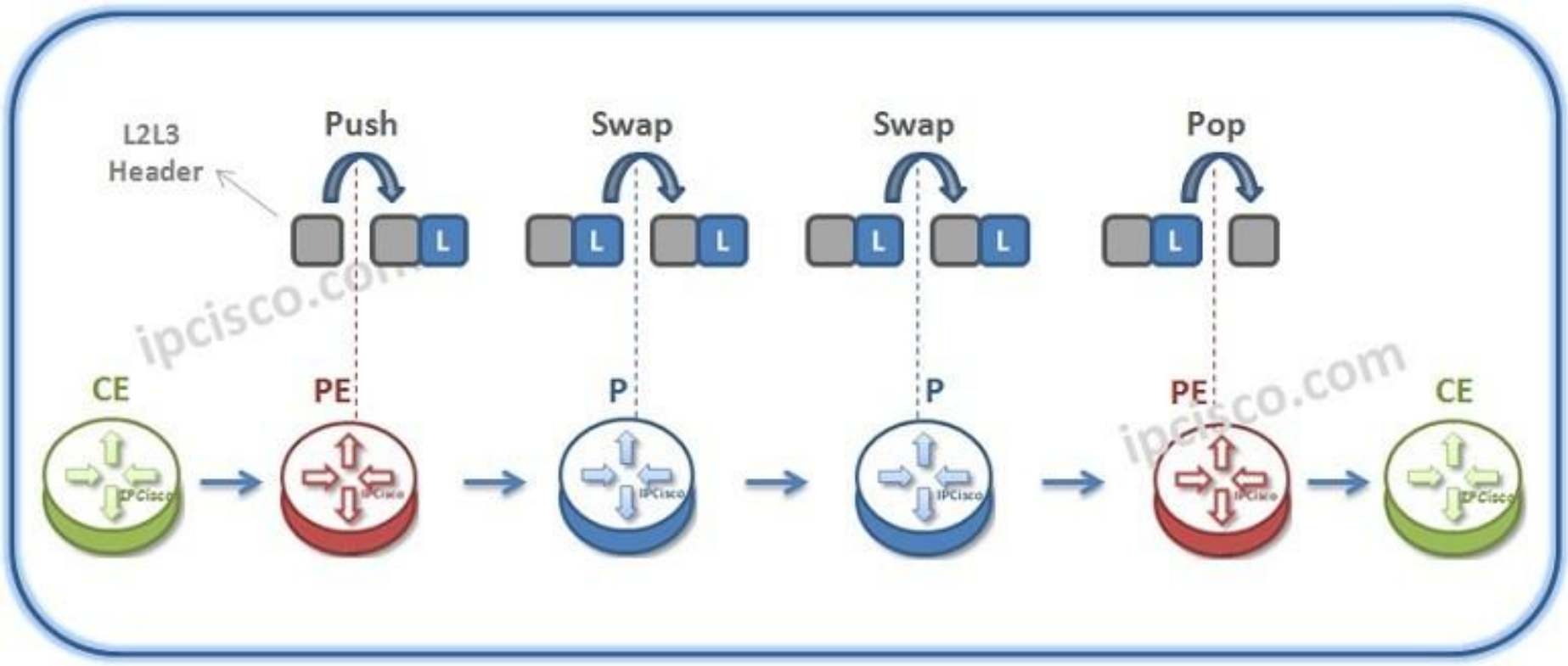
MPLS network consists of different router types used for switching and routing MPLS packets between end systems (hosts):

- **CE Routers (Customer Edge):** Located at the edge of the network, connecting to customer networks.
- **PE Routers (Provider Edge):** Located at the edge of the service provider's network.
- **P Routers (Provider):** Core routers within the service provider's network.

MPLS Operations: Push, Swap, and Pop



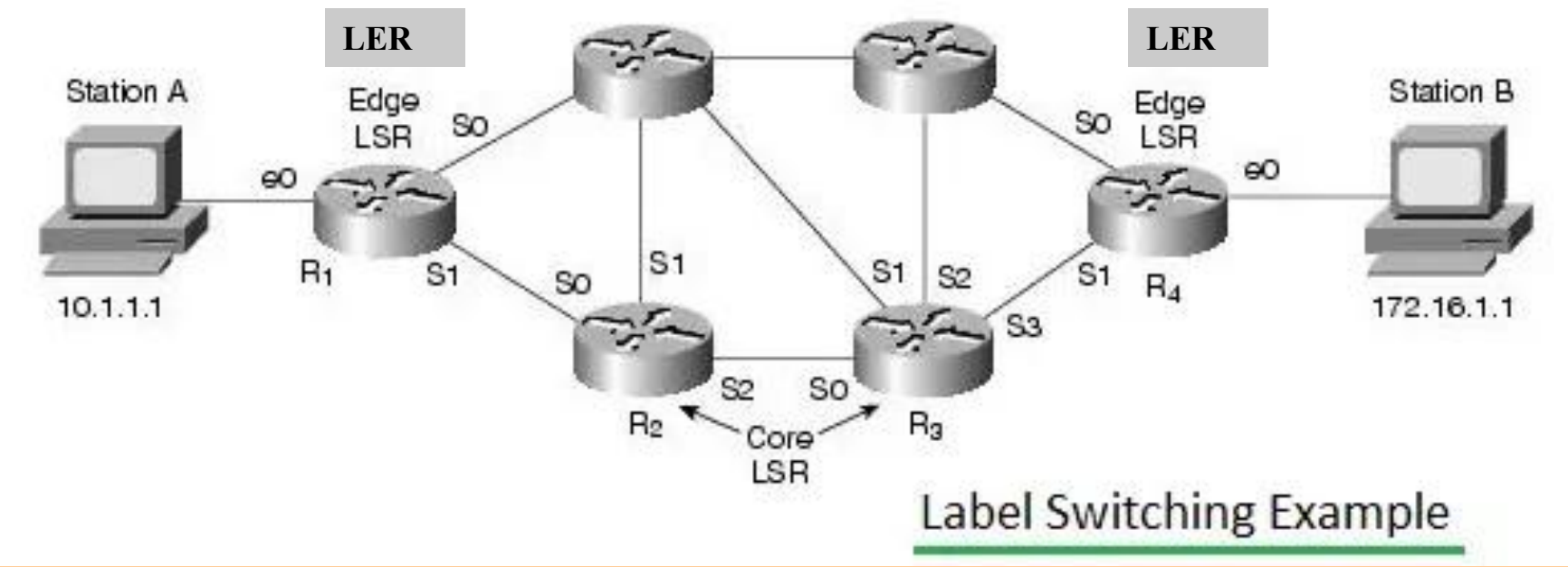
MPLS Operations: Push, Swap, and Pop



MPLS label switching relies on three basic operations:

- **Push:** The ingress (entry point) CE (Customer Edge) router assigns a label to the incoming packet.
- **Swap:** Routers within the Label Switched Path (LSP) swap the incoming label for an outgoing label.
- **Pop:** The egress (exit point) router removes the label from the packet before forwarding it to its final destination. This happens when the destination has been reached and the label is no longer needed.

Label Switching Example



label switching example between Station A and Station B:

Router	Incoming Label	Incoming Interface	Destination network	Outgoing interface	Outgoing Label
R1	**-**	**e0**	172.16.1	S1	6
R2	**6**	**S0**	172.16.1	**S2**	**11**
R3	**11**	**S0**	172.16.1	**S3**	**7**
R4	**7**	**S1**	172.16.1	**e0**	**-**

Router	Designation	Incoming Label	Incoming Interface	Destination network
R1	-	-	e0	172.16.1
R2		6	S0	172.16.1
R3		11	S0	172.16.1
R4		7	S1	172.16.1

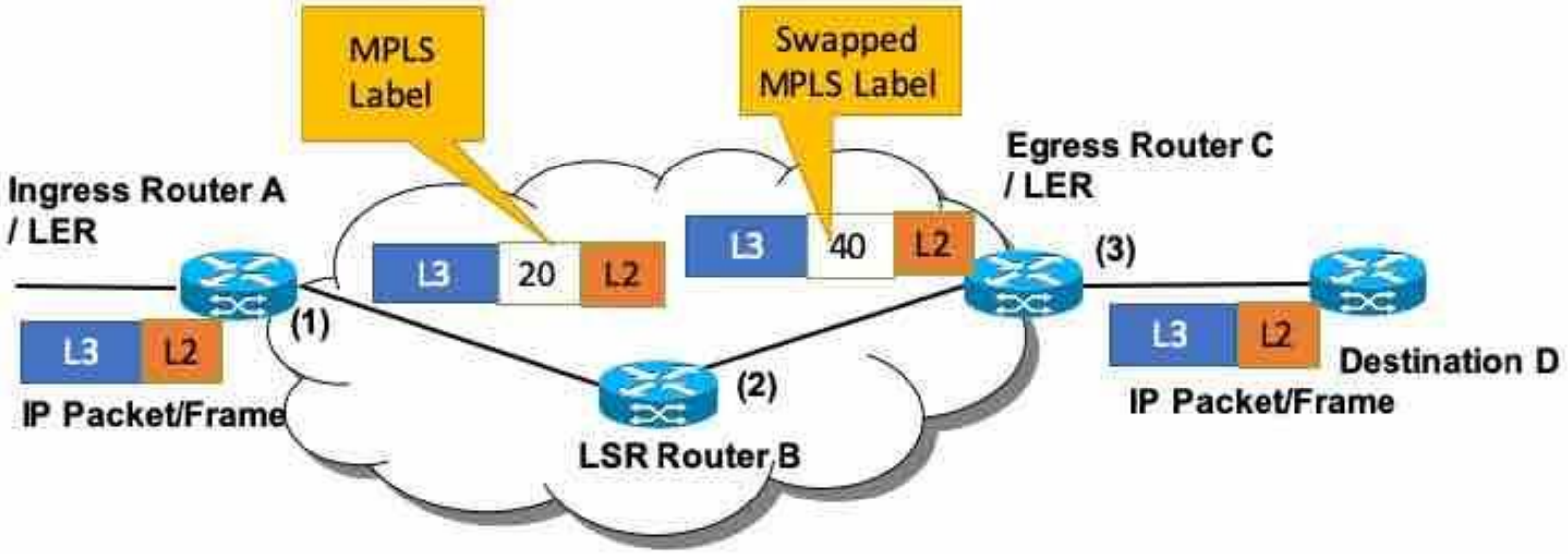
:oming Label	Incoming Interface	Destination network	Outgoing interface	Outgoing Label
	e0	172.16.1	S1	6
	S0	172.16.1	S2	11
	S0	172.16.1	S3	7
	S1	172.16.1	e0	-

Incoming Label	Outgoing Label	Outgoing Interface	Destination	Next Hop
-	20	B Intf (1)	D	B

Routing Table A

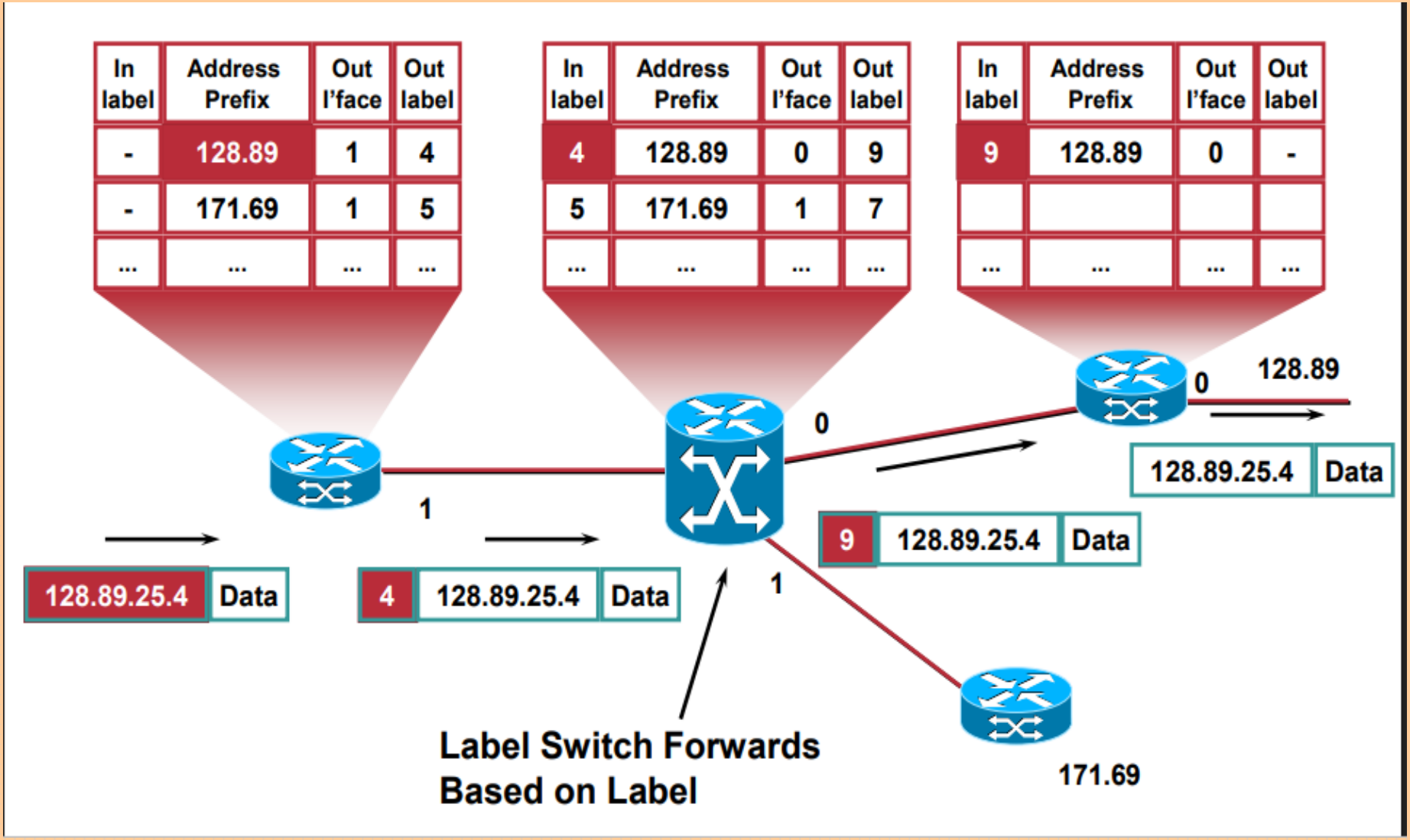
Incoming Label	Outgoing Label	Outgoing Interface	Destination	Next Hop
40	Pop	D Intf (3)	D	D

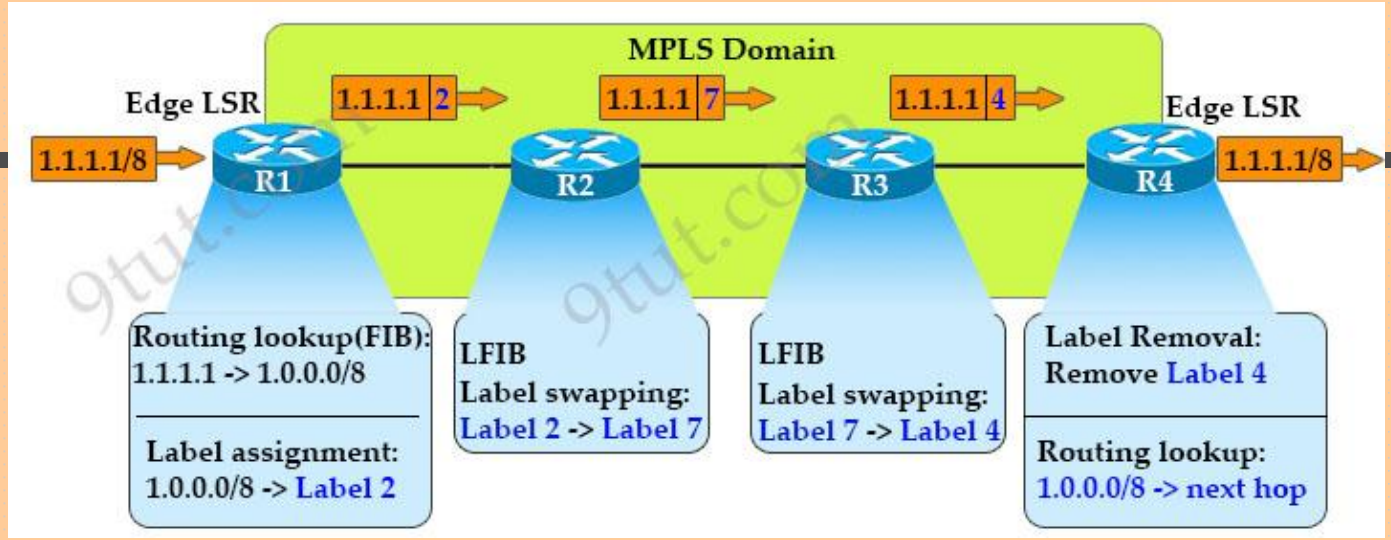
Routing Table C

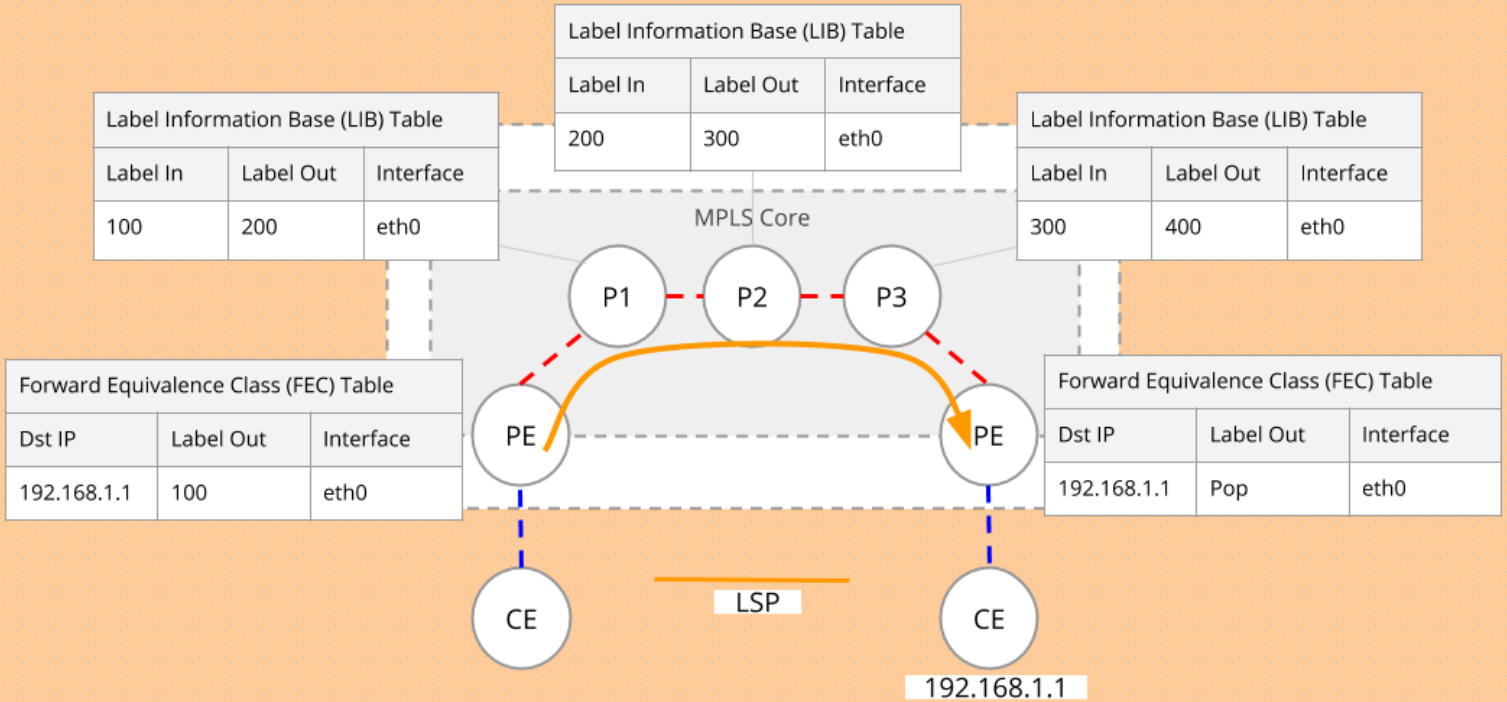


Incoming Label	Outgoing Interface	Outgoing Label	Destination	Next Hop
20	C Intf (2)	40	D	C

Routing Table B


In label
Address Prefix
Out l'face
Out label
-
128.89
1
4
-
171.69
1
5
...
...
...
...
In label
Address Prefix
Out l'face
Out label
4
128.89
0
9
5
171.69
1
7
...
...
...
...
In label
Address Prefix
Out l'face
Out label
9
128.89
0
-
...
...
...
...
128.89
0
0
1
1
128.89.25.4
Data
4
128.89.25.4
Data
9
128.89.25.4
Data
128.89.25.4
Data
Label Switch Forwards
Based on Label
171.69


MPLS Domain
Edge LSR
1.1.1.1 2
1.1.1.1 7
1.1.1.1 4
Edge LSR
1.1.1.1/8
R1
R2
R3
R4
1.1.1.1/8
Routing lookup(FIB):
1.1.1.1 -> 1.0.0.0/8
Label assignment:
1.0.0.0/8 -> Label 2
LFIB
Label swapping:
Label 2 -> Label 7
LFIB
Label swapping:
Label 7 -> Label 4
Label Removal:
Remove Label 4
Routing lookup:
1.0.0.0/8 -> next hop


Label Information Base (LIB) Table
Label In
Label Out
Interface
100
200
eth0
Label Information Base (LIB) Table
Label In
Label Out
Interface
200
300
eth0
Label Information Base (LIB) Table
Label In
Label Out
Interface
300
400
eth0
MPLS Core
P1
P2
P3
PE
PE
Forward Equivalence Class (FEC) Table
Dst IP
Label Out
Interface
192.168.1.1
100
eth0
Forward Equivalence Class (FEC) Table
Dst IP
Label Out
Interface
192.168.1.1
Pop
eth0
CE
LSP
CE
192.168.1.1

LERs (Label Edge Routers) and LSRs (Label Switching Routers) maintain tables:

- **FEC (Forwarding Equivalence Class) table:** {Label In, Destination IP Address, Label Out, Interface}
- **LIB (Label Information Base) table:** {Label In, Label Out, Interface }
- An LSR combines the functionalities of both a switch and a router.

Key Terminologies in MPLS

- **Provider Edge (PE) Router**: Router at the edge of the MPLS provider's network; adds/removes labels from packets.
- **Customer Edge (CE) Router**: Router at the customer's network edge that communicates with PE routers.
- **Label Switch Router (LSR)**: Router inside the MPLS core that understands and processes labels.
- **Ingress LSR**: First router that receives the packet from CE and pushes (adds) the MPLS header.
- **Intermediate LSR**: Routers that swap labels as packets move across the MPLS path.
- **Egress LSR**: The last router in the MPLS domain that pops (removes) the MPLS header before sending the packet to the CE.
- **Push, Pop, Swap**: Push -> Add a label, Pop -> Remove a label. & Swap -> Replace an existing label with a new one.

Working of MPLS (Forwarding Process)

- **Ingress Stage (Push)**: CE sends an IP packet to the PE (Ingress LSR). Ingress PE assigns a label based on the destination and attaches an MPLS header.
- **Core Stage (Swap)**: Intermediate LSRs forward the packet based only on the label. They swap the label with a new one as defined in their Label Forwarding Information Base (LFIB).
- **Egress Stage (Pop)**: The Egress PE removes the MPLS header and forwards the original IP packet to the CE router.

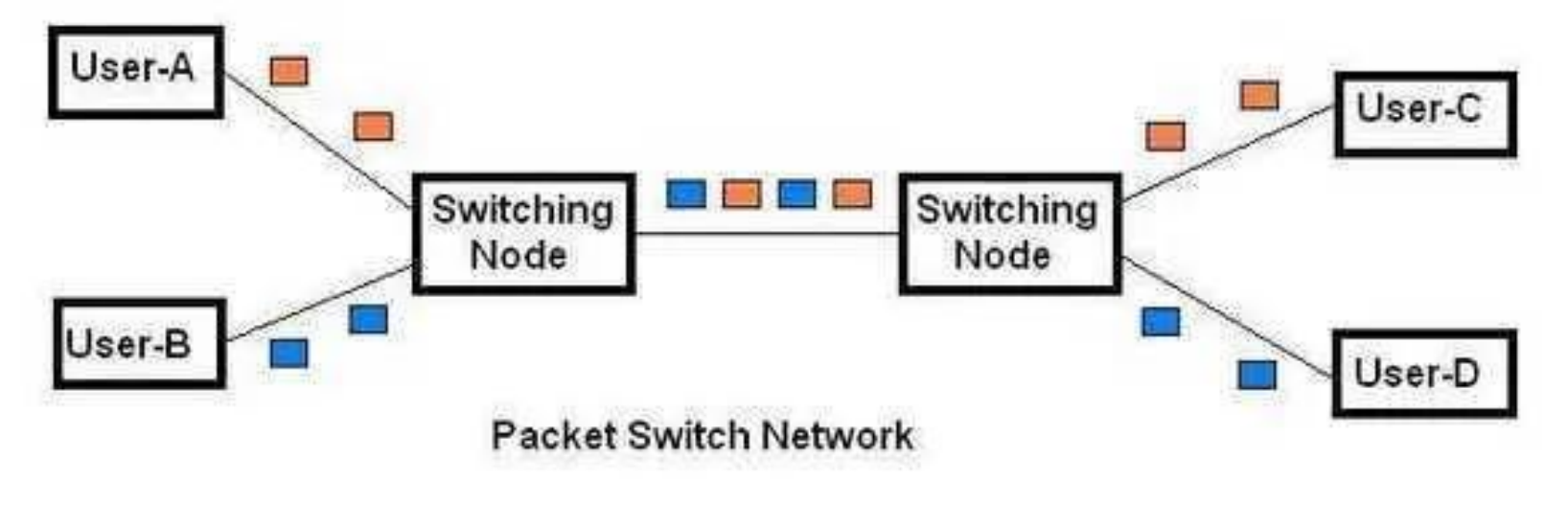
Benefits (Advantages) of MPLS Label Switching

- **Protocol Agnostic:** MPLS can forward any data across a network, not just IP traffic. It operates based on labels, not destination IP addresses.
- **Multi-Protocol Support:** It can transport various protocols, including ATM, SONET, and Ethernet.
- **Integrated Voice and Data:** Voice and data can be transported on the same network infrastructure.
- **Efficient Bandwidth Utilization:** MPLS can improve bandwidth utilization through traffic engineering.
- **Network Resiliency:** MPLS supports fast re-route mechanisms for improved network resilience.

Disadvantages of MPLS

- Expensive to implement compared to IP routing.
- Complexity in configuration and management.
- Security is weaker than encrypted VPN solutions.
- Less suitable for small-scale networks.
- Multiprotocol Label Switching (MPLS) Routing.

IP Packet Switching



Difference Between Label Switching (MPLS) and IP Switching

Feature

MPLS Label Switching

- ✓ Routes packet based on a 20-bit label.

IP Packet Switching

- ➢ Routes packet based on a 32-bit IP address.

Path Establishment

MPLS Label Switching

- ✓ Establishes a dedicated path known as an LSP (Label Switching Path) **before** data flow.

IP Packet Switching

- ➢ Doesn't establish a dedicated path. Data is transmitted as datagrams, routed based on IP addresses.

Forwarding Table

MPLS Label Switching

- ✓ Each router builds an LFIB (Label Forwarding Information Base) table using a protocol like LDP (Label Distribution Protocol).

IP Packet Switching

- ➢ Stores an IP routing table.

end